

第八章 分离变数法

本章基本要求

- 1. 熟练掌握分离变数法的基本思想与解题步骤；
- 2. 掌握求解含非齐次泛定方程的定解问题的几种方法；
- 3. 掌握求解含非齐次边界条件定解问题的基本方法。

§ 8.1 齐次方程的分离变数法

1. 长为 l 的弦，两端固定，弦中张力为 F_T . 在距一端为 x_0 的一点以力 F_0 把弦拉开，然后突然撤除这力（图 8-1）。求解弦的振动。

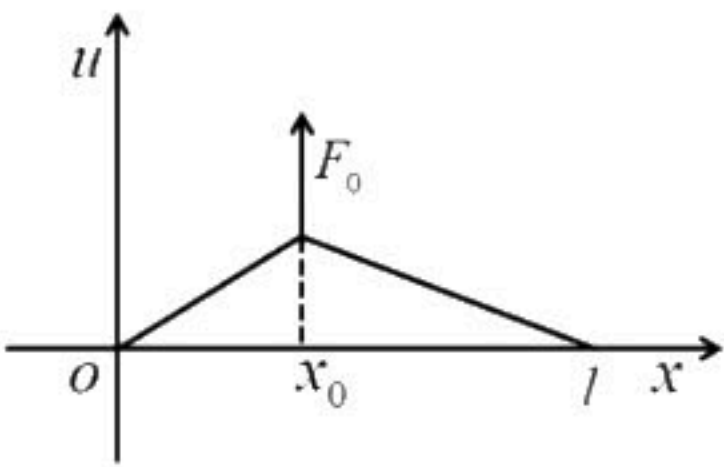


图8-1

解：定解问题为

$$u_{tt} - a^2 u_{xx} = 0 \tag{①}$$

$$u|_{x=0} = u|_{x=l} = 0 \tag{②}$$

$$u|_{t=0} = \begin{cases} \frac{F_0(l-x_0)}{F_T l} x, & (0 < x < x_0) \\ \frac{F_0 x_0}{F_T l} (l-x), & (x_0 < x < l) \end{cases} \tag{③}$$

$$u_t|_{t=0} = 0 \tag{④}$$

令 $u(x,t) = X(x)T(t)$ ，代入泛定方程①和边界条件②，并分离变数，得

$$\frac{T''}{a^2 T} = \frac{X''}{X} = -\lambda$$

这可分离为关于 X 的常微分方程和关于 T 的常微分方程，

$$\begin{cases} X'' + \lambda X = 0, \\ X(0) = 0, X(l) = 0; \end{cases} \tag{⑤}$$

$$T'' + \lambda a^2 T = 0 \tag{⑥}$$

本征问题⑤的本征值是 $\lambda_n = \frac{n^2 \pi^2}{l^2}, (n = 0, 1, 2, \dots)$ ，本征函数是

$$X_n(x) = \sin \frac{n\pi}{l} x, (n = 0, 1, 2, \dots)$$

方程⑥的解是

$$T_n(t) = A_n \cos \frac{n\pi a}{l} t + B_n \sin \frac{n\pi a}{l} t. \quad (n = 1, 2, \dots)$$

从而，问题的一般解是

$$u(x, t) = \sum_{n=1}^{\infty} \left(C_n \cos \frac{n\pi a t}{l} + D_n \sin \frac{n\pi a t}{l} \right) \sin \frac{n\pi x}{l}, \quad (7)$$

将⑦代入初值③ ④，有 $D_n = 0$ ，从而一般解⑦成为

$$u(x, t) = \sum_{n=1}^{\infty} C_n \cos \frac{n\pi a t}{l} \sin \frac{n\pi x}{l},$$

其中，

$$\begin{aligned} C_n &= \frac{2}{l} \left[\int_0^{x_0} \frac{F_0(l-x_0)}{F_T l} x \sin \frac{n\pi x}{l} dx + \int_{x_0}^l \frac{F_0 x_0}{F_T l} (l-x) \sin \frac{n\pi x}{l} dx \right] \\ &= \frac{2F_0 l}{F_T n^2 \pi^2} \sin \frac{n\pi x_0}{l} \end{aligned}$$

整理后，本题的解为

$$u(x, t) = \frac{2F_0 l}{F_T \pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi x}{l} \cos \frac{n\pi a t}{l}.$$

2. 求解细杆导热问题，杆长 l ， b 为常数，两端保持为零度，初始温度分布

$$u|_{t=0} = bx(l-x)/l^2。$$

解：定解问题为

$$u_t - a^2 u_{xx} = 0, \tag{①}$$

$$u|_{x=0} = u|_{x=l} = 0, \tag{②}$$

$$u|_{t=0} = bx(l-x)/l^2。 \tag{③}$$

令 $u(x,t) = X(x)T(t)$ ，代入泛定方程①和边界条件②，并分离变数，得

$$\frac{T'}{a^2 T} = \frac{X''}{X} = -\lambda$$

这可分离为关于 X 的常微分方程和关于 T 的常微分方程，

$$\begin{cases} X'' + \lambda X = 0, \\ X(0) = 0, X(l) = 0. \end{cases} \tag{④}$$

$$T' + \lambda a^2 T = 0 \tag{⑤}$$

本征问题④的本征值是 $\lambda_n = \frac{n^2 \pi^2}{l^2}, (n = 0, 1, 2, \dots)$ ，本征函数是

$$X_n(x) = \sin \frac{n\pi}{l} x, (n = 0, 1, 2, \dots)$$

方程⑤的解是

$$T_n(t) = C_n e^{-\frac{n^2 \pi^2 a^2 t}{l^2}}。$$

这样，问题的一般解为

$$u(x,t) = \sum_{n=1}^{\infty} C_n e^{-\frac{n^2 \pi^2 a^2 t}{l^2}} \sin \frac{n\pi x}{l} \tag{⑥}$$

将⑥代入初始条件③，有

$$u|_{t=0} = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi x}{l} = bx(l-x)/l^2。$$

系数

$$\begin{aligned}
C_n &= \frac{2}{l} \int_0^l \frac{b}{l^2} x(l-x) \sin \frac{n\pi x}{l} dx \\
&= -\frac{2b}{l^3} \frac{l}{n\pi} \int_0^l x(l-x) d\left(\cos \frac{n\pi x}{l}\right) \\
&= -\frac{2b}{n\pi l^2} \left[-\int_0^l (l-2x) \cos \frac{n\pi x}{l} dx \right] \\
&= \frac{2b}{n\pi l^2} \left[\frac{l}{n\pi} \int_0^l (l-2x) d\left(\sin \frac{n\pi x}{l}\right) \right] \\
&= \frac{2b}{n^2 \pi^2 l} \left(2 \int_0^l \sin \frac{n\pi x}{l} dx \right) \\
&= \frac{4b}{n^3 \pi^3} \cos \frac{n\pi x}{l} \Big|_l^0 \\
&= \frac{4b}{n^3 \pi^3} [1 - (-1)^n]
\end{aligned}$$

仅当 $n = 2k + 1$ 时, 该系数非零, 因而有

$$u(x, t) = \frac{8b}{\pi^3} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3} \sin \frac{(2k+1)\pi x}{l} e^{-\frac{(2k+1)^2 \pi^2 a^2 t}{l^2}}.$$

3. 两端固定弦，长为 l 。(1) 用宽为 2δ 的平面锤敲击弦 $x = x_0$ 点；(2) 用宽为 2δ 的余弦式凸面锤敲击弦的 $x = x_0$ 点。求解弦的振动。

解：

(1) 在敲打的瞬间，锤子给了该部分琴弦以均匀冲量，该部分琴弦获得了均匀速度 v_0 。因而，该定解问题为

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= 0 \\ u|_{x=0} &= u|_{x=l} = 0 \\ u|_{t=0} &= 0 \\ u_t|_{t=0} &= \begin{cases} v_0, & (x_0 - \delta < x < x_0 + \delta) \\ 0, & (\text{其他}) \end{cases} \end{aligned}$$

由于两端均为第一类齐次边值，可知，本征函数为 $\sin \frac{n\pi x}{l}$ ，从而一般解为

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi at}{l} + B_n \sin \frac{n\pi at}{l} \right) \sin \frac{n\pi x}{l}。$$

代入初始条件

$$\begin{aligned} \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} &= 0 \\ \sum_{n=1}^{\infty} \frac{n\pi a}{l} B_n \sin \frac{n\pi x}{l} &= \begin{cases} v_0, & (x_0 - \delta < x < x_0 + \delta) \\ 0, & (\text{other}) \end{cases} \end{aligned}$$

有， $A_n = 0$

$$\begin{aligned} \frac{n\pi a}{l} B_n &= \frac{2}{l} \left(\int_{x_0-\delta}^{x_0+\delta} v_0 \sin \frac{n\pi x}{l} dx \right) \\ &= \frac{2v_0}{l} \frac{l}{n\pi} \cos \frac{n\pi x}{l} \Big|_{x_0+\delta}^{x_0-\delta} \\ &= \frac{2v_0}{n\pi} \left(2 \sin \frac{n\pi x_0}{l} \sin \frac{n\pi \delta}{l} \right) \\ B_n &= \frac{4lv_0}{n^2 \pi^2 a} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi \delta}{l} \end{aligned}$$

从而，其解为

$$u(x, t) = \frac{4v_0 l}{\pi^2 a} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi \delta}{l} \sin \frac{n\pi at}{l} \sin \frac{n\pi x}{l}。$$

(2) 在敲打的瞬间, 锤子给了该部分琴弦以余弦形式的冲量, 该部分琴弦获得了速度 $v_0 \cos \frac{x-x_0}{2\delta}$, 因而, 该定解问题为

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= 0 \\ u|_{x=0} &= u|_{x=l} = 0 \\ u|_{t=0} &= 0 \\ u_t|_{t=0} &= \begin{cases} v_0 \cos \frac{\pi(x-x_0)}{2\delta}, & (x_0 - \delta < x < x_0 + \delta) \\ 0, & (\text{其他}) \end{cases} \end{aligned}$$

其一般解为

$$u(x, t) = \sum_{n=0}^{\infty} B_n \sin \frac{n\pi at}{l} \sin \frac{n\pi x}{l}。$$

其中, 系数 B_n 由初值决定

$$\begin{aligned} \frac{n\pi a}{l} B_n &= \frac{2}{l} \left[\int_{x_0-\delta}^{x_0+\delta} v_0 \cos \frac{\pi(x-x_0)}{2\delta} \sin \frac{n\pi x}{l} dx \right] \\ &= \frac{v_0}{l} \int_{x_0-\delta}^{x_0+\delta} \left\{ \sin \left[\left(\frac{n\pi}{l} + \frac{\pi}{2\delta} \right) x - \frac{\pi x_0}{2\delta} \right] + \sin \left[\left(\frac{n\pi}{l} - \frac{\pi}{2\delta} \right) x + \frac{\pi x_0}{2\delta} \right] \right\} dx \\ &= \frac{v_0}{\pi l} \left\{ \frac{2\delta l}{2n\delta + l} \cos \left[\left(\frac{n\pi}{l} + \frac{\pi}{2\delta} \right) x - \frac{\pi x_0}{2\delta} \right] + \frac{2\delta l}{2n\delta - l} \cos \left[\left(\frac{n\pi}{l} - \frac{\pi}{2\delta} \right) x + \frac{\pi x_0}{2\delta} \right] \right\}_{x_0-\delta}^{x_0+\delta} \\ &= \frac{v_0}{\pi l} \cdot \frac{4\delta l^2}{(2n\delta)^2 - l^2} \left[-\sin \frac{n\pi(x_0-\delta)}{l} - \sin \frac{n\pi(x_0+\delta)}{l} \right] \\ &= -\frac{v_0}{\pi} \cdot \frac{8\delta l}{(2n\delta)^2 - l^2} \sin \frac{n\pi x_0}{l} \cos \frac{n\pi \delta}{l}。 \end{aligned}$$

$$\begin{aligned} B_n &= -\frac{v_0}{n\pi^2} \cdot \frac{8\delta l^2}{(2n\delta)^2 - l^2} \sin \frac{n\pi x_0}{l} \cos \frac{n\pi \delta}{l} \\ &= \frac{8v_0 \delta}{\pi^2 a} \cdot \frac{1}{n} \cdot \frac{1}{1 - \left(\frac{2\delta n}{l} \right)^2} \sin \frac{n\pi x_0}{l} \cos \frac{n\pi \delta}{l}。 \end{aligned}$$

从而, 最终解为

$$u(x, t) = \frac{8v_0 \delta}{\pi^2 a} \sum_{n=1}^{\infty} \frac{1}{n} \frac{1}{1 - \left(\frac{2\delta n}{l} \right)^2} \sin \frac{n\pi x_0}{l} \cos \frac{n\pi \delta}{l} \sin \frac{n\pi at}{l} \sin \frac{n\pi x}{l}。$$

4. 长为 l 的均匀杆，两端受压从而长度缩为 $l(1-2\varepsilon)$ 。放手后自由振动，求解杆的纵振动。

解：参见图 8-2，初始时刻，杆的中心没有位移，其余各点的位移为

$$\frac{x-l/2}{l/2}(-\varepsilon l)=\varepsilon(l-2x), \text{ 定解问题为}$$

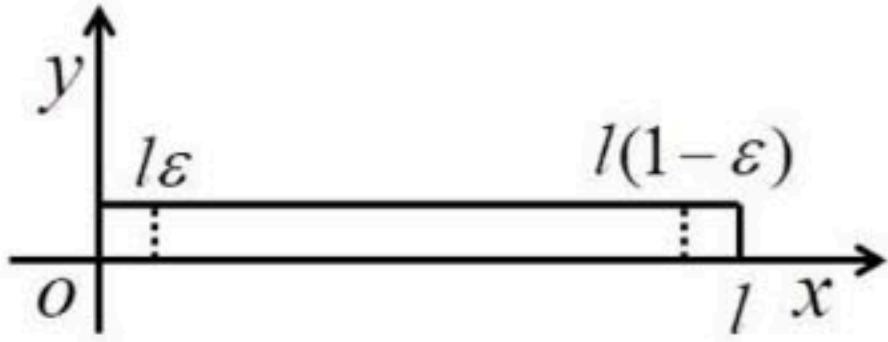


图8-2

$$\begin{aligned} u_{tt}-a^2u_{xx}&=0 \\ u_x|_{x=0}&=u_x|_{x=l}=0, \\ u|_{t=0}&=\varepsilon(l-2x), \quad u_t|_{t=0}=0. \end{aligned}$$

由于两端均为第二类齐次边值，可知，本征函数为 $\cos\frac{n\pi x}{l}$ ，从而一般解为

$$u(x,t)=C_0+\sum_{n=1}^{\infty}\left(C_n\cos\frac{n\pi at}{l}+D_n\sin\frac{n\pi at}{l}\right)\cos\frac{n\pi x}{l}.$$

将此解代入初始条件

$$\begin{aligned} C_0+\sum_{n=1}^{\infty}C_n\cos\frac{n\pi x}{l}&=\varepsilon(l-2x) \\ \sum_{n=0}^{\infty}\frac{n\pi a}{l}D_n\cos\frac{n\pi x}{l}&=0 \end{aligned}$$

有 $D_n=0$,

$$\begin{aligned} C_0&=\frac{1}{l}\int_0^l\varepsilon(l-2x)\mathrm{d}x=0 \\ C_n&=\frac{2}{l}\int_0^l\varepsilon(l-2x)\cos\frac{n\pi x}{l}\mathrm{d}x \\ &=\frac{2}{l}\frac{\varepsilon l}{n\pi}\int_0^l(l-2x)\mathrm{d}\left(\sin\frac{n\pi x}{l}\right) \\ &=\frac{2}{l}\frac{\varepsilon l}{n\pi}\left(2\int_0^l\sin\frac{n\pi x}{l}\mathrm{d}x\right) \\ &=\frac{4\varepsilon l}{n^2\pi^2}\left[1-(-1)^n\right]=\begin{cases}\frac{8\varepsilon l}{(2k+1)^2\pi^2} & (n=2k+1, \quad k=0,1,2,\dots) \\ 0 & (n=2k, \quad k=0,1,2,\dots) \end{cases} \end{aligned}$$

从而，解为

$$u(x,t)=\frac{8\varepsilon l}{\pi^2}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2}\cos\frac{(2k+1)\pi at}{l}\cos\frac{(2k+1)\pi x}{l}.$$

$$u(x,t)=\frac{8\varepsilon l}{\pi^2}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2}\cos\frac{(2k+1)\pi at}{l}\cos\frac{(2k+1)\pi x}{l}.$$

5. 长为 l 的杆，一端固定，另一端受力 F_0 而伸长（图 8-3）。求解杆在放手后的振动。

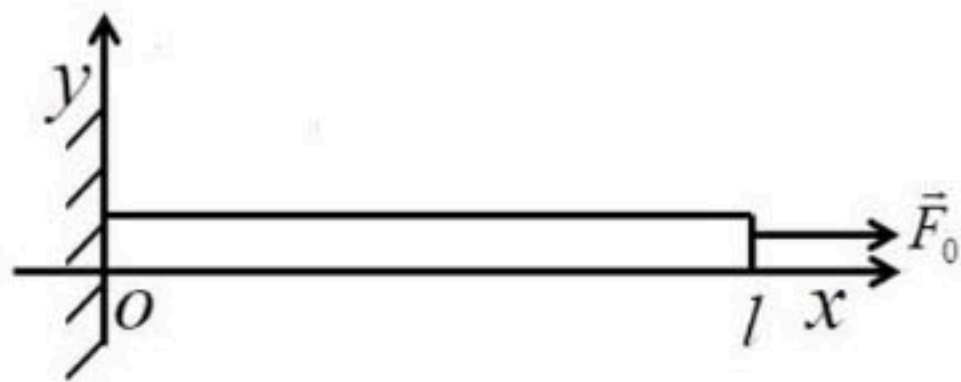


图8-3

解：初始时刻，杆受力平衡，处处均有应力 F_0 ，因而处处形变为 $u_x = F_0/ES$ 。据此，初始位

移为 $u|_{t=0} = \int_0^x (F_0/ES) d\xi = F_0 x/ES$ 。定解问题为

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= 0 \\ u|_{x=0} &= 0, \quad u_x|_{x=l} = 0, \\ u|_{t=0} &= \frac{F_0}{ES} x, \quad u_t|_{t=0} = 0. \end{aligned}$$

本题边界条件，左端为第一类齐次，右端为第二类齐次，与教材本节例 2 边值相同，从而其本证函

数为 $\sin \frac{(2n+1)\pi x}{2l}$ ，问题的一般解为

$$u(x,t) = \sum_{n=0}^{\infty} \left[C_n \cos \frac{(2n+1)\pi at}{2l} + D_n \sin \frac{(2n+1)\pi at}{2l} \right] \sin \frac{(2n+1)\pi x}{2l}.$$

将此解代入初值

$$\begin{aligned} \sum_{n=0}^{\infty} C_n \sin \frac{(2n+1)\pi x}{2l} &= \frac{F_0}{ES} x \\ \sum_{n=0}^{\infty} \frac{(2n+1)\pi a}{2l} D_n \sin \frac{(2n+1)\pi x}{2l} &= 0 \end{aligned}$$

有

$$\begin{aligned} D_n &= 0 \\ C_n &= \frac{2}{l} \int_0^l \frac{F_0}{ES} x \sin \frac{(2n+1)\pi x}{2l} dx \\ &= -\frac{2F_0}{ESl} \frac{2l}{(2n+1)\pi} \int_0^l x d \left[\cos \frac{(2n+1)\pi x}{2l} \right] \\ &= \frac{2F_0}{ESl} \frac{2l}{(2n+1)\pi} \int_0^l \cos \frac{(2n+1)\pi x}{2l} dx \\ &= (-1)^n \frac{8lF_0}{(2n+1)^2 \pi^2 ES} \end{aligned}$$

从而解为

$$u(x,t) = \frac{8lF_0}{\pi^2 ES} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)^2} \cos \frac{(2n+1)\pi at}{2l} \sin \frac{(2n+1)\pi x}{2l}.$$

6. 长为 l 的理想传输线, 远端开路. 先把传输线充电到电位差 v_0 , 然后把近端短路. 求解线上电压 $v(x, t)$ 。

解: 远端开路, 电流为零, 因而 $x=l$ 的边界条件为 $v_x|_{x=l}=0$, 近端 $x=0$ 短路, 其电压降为零, 因而 $v|_{x=0}=0$ 。初始时刻, 电位差为 $v|_{t=0}=v_0$, 由于初始时刻, 电流为零, $v_t|_{t=0}=0$ 。因而定解问题为

$$\begin{aligned} v_{tt} - a^2 v_{xx} &= 0 \\ v|_{x=0} &= 0, \quad v_x|_{x=l} = -L \frac{d}{dt}(j|_{x=l}) = 0, \\ v|_{t=0} &= v_0, \quad v_t|_{t=0} = -\frac{1}{C} j_x|_{t=0} = 0. \end{aligned}$$

本题与第 5 题具有相同边界条件, 从而其一般解为

$$u(x, t) = \sum_{n=0}^{\infty} \left[C_n \cos \frac{(2n+1)\pi at}{2l} + D_n \sin \frac{(2n+1)\pi at}{2l} \right] \sin \frac{(2n+1)\pi x}{2l},$$

将此解代入初值

$$\begin{aligned} \sum_{n=0}^{\infty} C_n \sin \frac{(2n+1)\pi x}{2l} &= v_0 \\ \sum_{n=0}^{\infty} \frac{(2n+1)\pi a}{2l} D_n \sin \frac{(2n+1)\pi x}{2l} &= 0 \end{aligned}$$

有

$$\begin{aligned} D_n &= 0 \\ C_n &= \frac{2}{l} \int_0^l v_0 \sin \frac{(2n+1)\pi x}{2l} dx \\ &= \frac{2v_0}{l} \frac{2l}{(2n+1)\pi} \cos \frac{(2n+1)\pi x}{2l} \Big|_l^0 \\ &= \frac{4v_0}{(2n+1)\pi} \end{aligned}$$

从而, 问题的解为

$$v(x, t) = \frac{4v_0}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} \cos \frac{(2n+1)\pi at}{2l} \sin \frac{(2n+1)\pi x}{2l}.$$

7. 长为 l 的杆，上端固定在电梯天花板，杆身竖直，下端自由. 电梯下降，当速度为 v_0 时突然停止。求解杆的振动。

解：杆的边界条件为 $u|_{x=0} = 0$, $u_x|_{x=l} = 0$ ，初始位移为零，初始速度为 v_0 ，因而定解问题为

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= 0 \\ u|_{x=0} &= 0, \quad u_x|_{x=l} = 0 \\ u|_{t=0} &= 0, \quad u_t|_{t=0} = v_0 \end{aligned}$$

本题与第 5 题具有相同边界条件，从而其一般解为

$$u(x, t) = \sum_{n=0}^{\infty} \left[C_n \cos \frac{(2n+1)\pi at}{2l} + D_n \sin \frac{(2n+1)\pi at}{2l} \right] \sin \frac{(2n+1)\pi x}{2l},$$

将此解代入初值

$$\begin{aligned} \sum_{n=0}^{\infty} C_n \sin \frac{(2n+1)\pi x}{2l} &= 0 \\ \sum_{n=0}^{\infty} \frac{(2n+1)\pi a}{2l} D_n \sin \frac{(2n+1)\pi x}{2l} &= v_0 \end{aligned}$$

其中，

$$\begin{aligned} C_n &= 0 \\ D_n &= \frac{2l}{(2n+1)\pi a} \frac{2}{l} \int_0^l v_0 \sin \frac{(2n+1)\pi x}{2l} dx \\ &= \frac{2v_0}{l} \frac{4l^2}{(2n+1)^2 \pi^2 a} \cos \frac{(2n+1)\pi x}{2l} \Big|_l^0 \\ &= \frac{8v_0 l}{(2n+1)^2 \pi^2 a} \end{aligned}$$

因而，最终解为

$$u(x, t) = \frac{8v_0 l}{\pi^2 a} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \sin \frac{(2n+1)\pi at}{2l} \sin \frac{(2n+1)\pi x}{2l}.$$

8. 在铀块中,除了中子的扩散运动以外,还进行着中子的增殖过程,每秒钟在单位体积中产生的中子数正比于该处的中子浓度 u ,从而可表为 βu (β 是表示增殖快慢的常数)。研究厚度为 l 的层状铀块。求临界厚度。(铀块厚度超过临界厚度,则中子浓度将随着时间而急剧增长以致铀块爆炸,原子弹里就是这么回事)

解:考虑沿铀块厚度方向的一维问题(图 8-4),其扩散方程为

$$u_t - a^2 u_{xx} = \beta u, \quad a^2 = D, \tag{①}$$

解法一 令 $u(x,t) = X(x)T(t)$, 代入方程①,有

$$XT' - a^2 X''T - \beta XT = 0,$$

分离变数

$$\frac{T' - \beta T}{a^2 T} = \frac{X''}{X} = -\lambda^2。$$

有

$$\begin{cases} X'' + \lambda^2 X = 0, \\ X(0) = X(l) = 0 \end{cases} \tag{②}$$

$$T' + (a^2 \lambda^2 - \beta)T = 0。 \tag{③}$$

本证问题②的本征函数与本征值分别是 $\sin \frac{n\pi x}{l}$, $\lambda^2 = \frac{n^2 \pi^2}{l^2}$ 。方程③的解是

$$T_n = C_n e^{-\left(\frac{n^2 \pi^2 a^2}{l^2} - \beta\right)t}$$

从而,方程①的解是

$$u(x,t) = \sum_{n=1}^{\infty} C_n e^{-\left(\frac{n^2 \pi^2 a^2}{l^2} - \beta\right)t} \sin \frac{n\pi x}{l}。$$

该解的行为随指数不同分为三种情形:

- i) $\frac{n^2 \pi^2 a^2}{l^2} - \beta > 0$, 浓度随时间减少, 反应熄灭;
- ii) $\frac{n^2 \pi^2 a^2}{l^2} - \beta < 0$, 浓度随时间增长, 发生爆炸;
- iii) $\frac{n^2 \pi^2 a^2}{l^2} - \beta = 0$, 浓度不随时间变化, 此时厚度即为临界度。取 $n=1$,得临界厚度为(记 l 为 L)

$$L = a\pi / \sqrt{\beta}。$$

解法二 由解法一知,达到临界厚度时,中子浓度不随时间变化,亦即 $u_t = 0$ 。因而,方程①成为

$$\frac{d^2 u}{dx^2} + \frac{\beta}{a^2} u = 0,$$

解得

$$u(x) = A \cos \frac{\sqrt{\beta}}{a} x + B \sin \frac{\sqrt{\beta}}{a} x。$$

代入边界条件 $u|_{x=0} = u|_{x=l} = 0$, 有

$$\begin{cases} A = 0 \\ B \sin \frac{\sqrt{\beta}}{a} l = 0 \end{cases}$$

要使得解非零,有

$$\frac{\sqrt{\beta}}{a} l = \pi,$$

因而,临界厚度为

$$L = \pi a / \sqrt{\beta}。$$

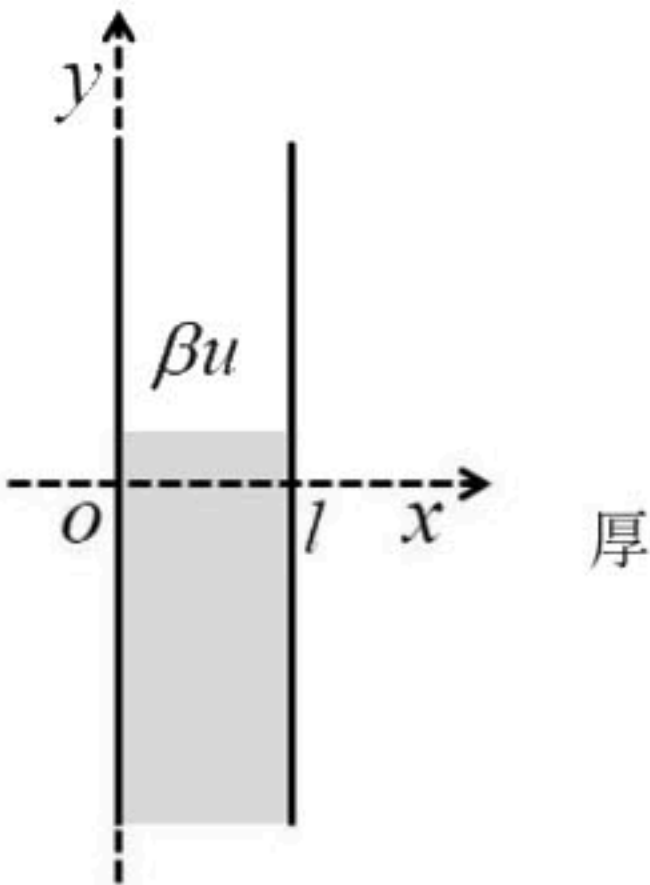


图8-4 化,

解得

$$u(x) = A \cos \frac{\sqrt{\beta}}{a} x + B \sin \frac{\sqrt{\beta}}{a} x。$$

代入边界条件 $u|_{x=0} = u|_{x=l} = 0$ ，有

$$\begin{cases} A = 0 \\ B \sin \frac{\sqrt{\beta}}{a} l = 0 \end{cases}$$

要使得解非零，有

$$\frac{\sqrt{\beta}}{a} l = \pi，$$

因而，临界厚度为

$$L = \pi a / \sqrt{\beta}。$$

9. 求解薄膜的恒定表面浓度扩散问题. 薄膜厚度为 l , 杂质从两面进入薄膜. 由于薄膜周围气氛中含有充分的杂质, 薄膜表面上的杂质浓度得以保持为恒定的 N_0 , 对于较大的 t , 把所得答案简化。

解：该定解问题可以看作一维扩散问题

$$\begin{aligned} u_t - a^2 u_{xx} &= 0 \\ u|_{x=0} &= N_0, \quad u|_{x=l} = N_0 \\ u|_{t=0} &= 0 \end{aligned}$$

该定解问题的边界条件为第一类非齐次边界条件，可以通过 $u = w + N_0$ ，将该边界条件齐次化，原定解问题转换成 w 的定解问题

$$\begin{aligned} w_t - a^2 w_{xx} &= 0 \\ w|_{x=0} &= 0, \quad w|_{x=l} = 0 \\ w|_{t=0} &= -N_0 \end{aligned}$$

与第 2 题类似，其一般解为

$$w(x,t) = \sum_{n=0}^{\infty} D_n e^{-\frac{n^2 \pi^2 a^2 t}{l^2}} \sin \frac{n \pi x}{l}。$$

将此解代入初值，有

$$\sum_{n=0}^{\infty} D_n e^{-\frac{n^2 \pi^2 a^2 t}{l^2}} \sin \frac{n \pi x}{l} = -N_0，$$

其中

$$\begin{aligned} D_n &= -\frac{2}{l} \int_0^l N_0 \sin \frac{n \pi x}{l} dx = \frac{2 N_0}{n \pi} [(-1)^n - 1] \\ &= \begin{cases} 0 & (n=2k) \\ \frac{2 N_0}{(2k+1) \pi} & (n=2k+1) \end{cases} \end{aligned}$$

其中， $k = 0, 1, 2, \dots$ 。因而，原定解问题的解为

$$u(x,t) = N_0 - \frac{4 N_0}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} e^{-\frac{(2k+1)^2 \pi^2 a^2 t}{l^2}} \sin \frac{(2k+1) \pi x}{l}。$$

当 t 足够大时，保留主要的衰减项，有

$$u(x,t) \approx N_0 - \frac{4 N_0}{\pi} e^{-\frac{\pi^2 a^2 t}{l^2}} \sin \frac{\pi x}{l}。 \text{ (图 8-5)}$$

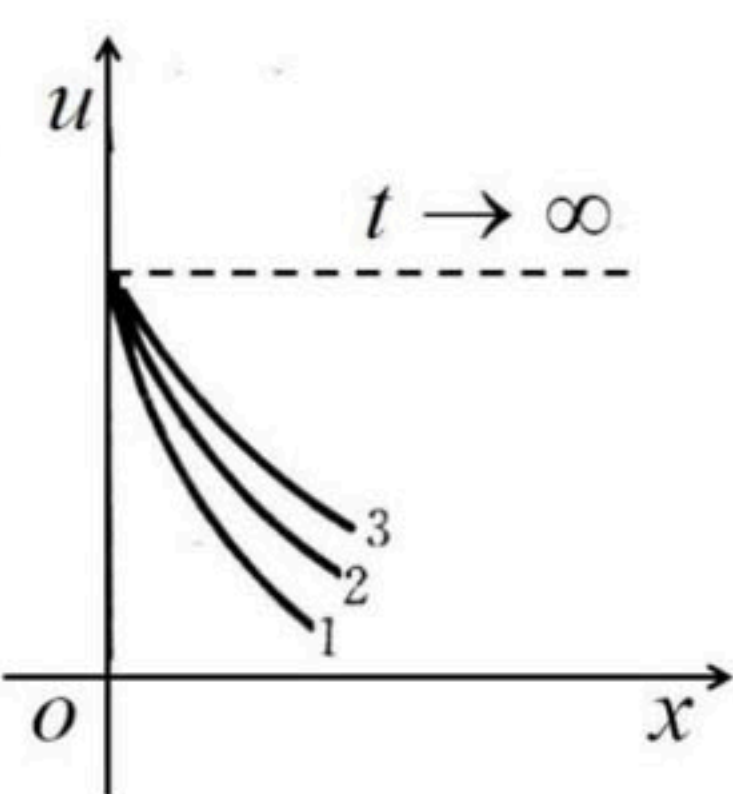


图8-5

10. 把上题改为限定源扩散。这是说，薄膜两面的表层已含有一定的杂质，比方说，每单位表面积下杂质总量 Φ_0 ，但此外不再有杂质进入薄膜。

解：没有杂质源时，体系不再对外有杂质交换，因而边界处浓度梯度均为零。考虑到初始时刻，杂质都集中在薄膜的两个表面，因而，上述定解问题为

$$\begin{aligned} u_t - a^2 u_{xx} &= 0 \\ u_x|_{x=0} &= 0, \quad u_x|_{x=l} = 0 \\ u|_{t=0} &= \Phi_0 \delta(x) + \Phi_0 \delta(x-l) \end{aligned}$$

其两端为第二类齐次边值，一般解为

$$u(x,t) = C_0 + \sum_{n=1}^{\infty} C_n e^{-\frac{n^2 \pi^2 a^2 t}{l^2}} \cos \frac{n \pi x}{l}$$

将此解代入初值

$$C_0 + \sum_{n=1}^{\infty} C_n \cos \frac{n \pi x}{l} = \Phi_0 \delta(x) + \Phi_0 \delta(x-l)$$

其中，

$$\begin{aligned} C_0 &= \frac{1}{l} \int_0^l [\Phi_0 \delta(x) + \Phi_0 \delta(x-l)] dx = \frac{2\Phi_0}{l} \\ C_n &= \frac{2}{l} \int_0^l [\Phi_0 \delta(x) + \Phi_0 \delta(x-l)] \cos \frac{n \pi x}{l} dx = \frac{2\Phi_0}{l} [1 + (-1)^n] \\ &= \begin{cases} \frac{4\Phi_0}{l}, & (n = 2k, \quad k = 1, 2, \dots) \\ 0, & (n = 2k + 1, \quad k = 1, 2, \dots) \end{cases} \end{aligned}$$

因而，解为

$$u(x,t) = \frac{4\Phi_0}{l} \sum_{k=0}^{\infty} \frac{1}{\delta_k} e^{-\frac{4k^2 \pi^2 a^2 t}{l^2}} \cos \frac{2k \pi x}{l}。$$

当 t 足够大时，保留主要的衰减项，有

$$u(x,t) \approx \frac{2\Phi_0}{l} + \frac{4\Phi_0}{l} e^{-\frac{4\pi^2 a^2 t}{l^2}} \cos \frac{2\pi x}{l}。 \quad (\text{图 8-6})$$

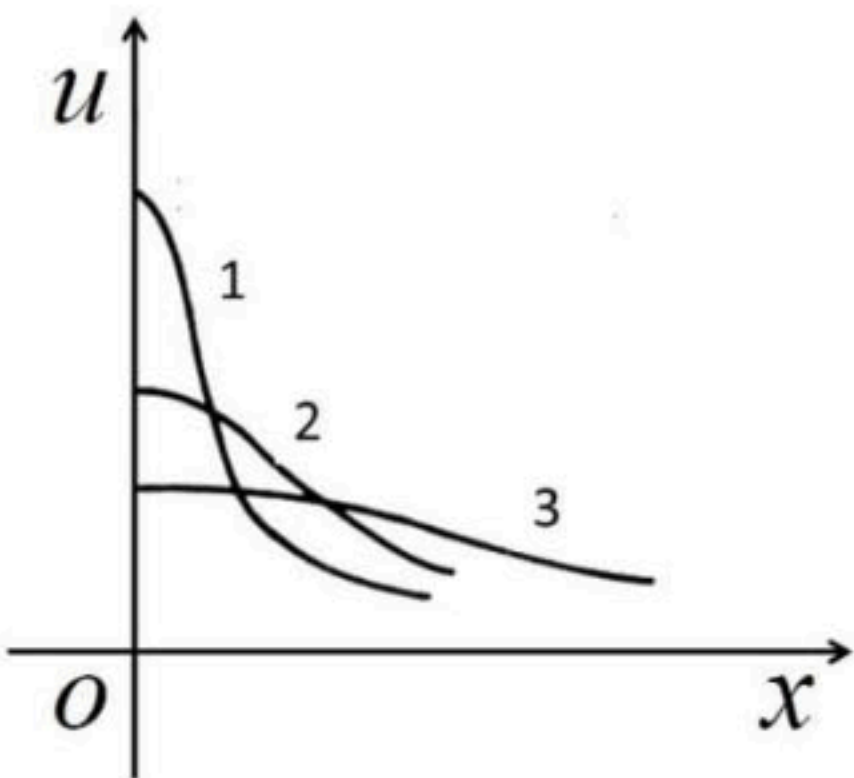


图8-6

11. 在矩形区域 $0 < x < a, 0 < y < b$ 上求解拉普拉斯方程 $\Delta u = 0$ ，使满足边界条件；

$$u|_{x=0} = Ay(b-y), u|_{x=a} = 0, u|_{y=0} = B \sin \frac{\pi x}{a}, u|_{y=b} = 0。$$

解：令 $u = w + v$ ，将原定解问题分成 w 和 v 两个定解问题

$$\begin{cases} w_{xx} + w_{yy} = 0 \\ w|_{x=0} = 0, & w|_{x=a} = 0 \\ w|_{y=0} = B \sin \frac{\pi x}{a}, & w|_{y=b} = 0 \end{cases} \quad \begin{cases} v_{xx} + v_{yy} = 0 \\ v|_{x=0} = Ay(b-y), & v|_{x=a} = 0 \\ v|_{y=0} = 0, & v|_{y=b} = 0 \end{cases}$$

对于 w 的定解问题，有

$$w(x,y) = \sum_{n=1}^{\infty} \sin \frac{n\pi x}{a} \left(C_n e^{\frac{n\pi y}{a}} + D_n e^{-\frac{n\pi y}{a}} \right),$$

代入 y 方向边值

$$\begin{aligned} \sum_{n=1}^{\infty} \sin \frac{n\pi x}{a} (C_n + D_n) &= B \sin \frac{\pi x}{a} \\ \sum_{n=1}^{\infty} \sin \frac{n\pi x}{a} \left(C_n e^{\frac{n\pi b}{a}} + D_n e^{-\frac{n\pi b}{a}} \right) &= 0 \end{aligned}$$

有

$$\begin{cases} C_1 + D_1 = B \\ C_1 e^{\frac{\pi b}{a}} + D_1 e^{-\frac{\pi b}{a}} = 0 \end{cases} \quad \begin{cases} C_{n \neq 1} + D_{n \neq 1} = 0 \\ C_{n \neq 1} e^{\frac{n\pi b}{a}} + D_{n \neq 1} e^{-\frac{n\pi b}{a}} = 0 \end{cases}$$

解得

$$\begin{cases} C_1 = \frac{B}{1 - e^{\frac{2\pi b}{a}}} = -\frac{1}{2} \frac{B e^{\frac{\pi b}{a}}}{\operatorname{sh} \frac{\pi b}{a}} \\ D_1 = -\frac{e^{\frac{2\pi b}{a}} B}{1 - e^{\frac{2\pi b}{a}}} = \frac{1}{2} \frac{B e^{\frac{\pi b}{a}}}{\operatorname{sh} \frac{\pi b}{a}} \\ C_{n \neq 1} = D_{n \neq 1} = 0 \end{cases}$$

w 的解为

$$w(x,y) = \sin \frac{\pi x}{a} \left[-\frac{1}{2} \frac{B e^{\frac{\pi(y-b)}{a}}}{\operatorname{sh} \frac{\pi b}{a}} + \frac{1}{2} \frac{B e^{-\frac{\pi(y-b)}{a}}}{\operatorname{sh} \frac{\pi b}{a}} \right] = B \sin \frac{\pi x}{a} \frac{\operatorname{sh} \frac{\pi(b-y)}{a}}{\operatorname{sh} \frac{\pi b}{a}}。$$

对于 v 的定解问题，其一般解为

$$v(x,y)=\sum_{n=1}^{\infty}\sin\frac{n\pi y}{b}\left(C_n\mathrm{e}^{\frac{n\pi x}{b}}+D_n\mathrm{e}^{-\frac{n\pi x}{b}}\right)。$$

代入 x 方向边值

$$\begin{aligned}\sum_{n=1}^{\infty}\sin\frac{n\pi y}{b}(C_n+D_n)&=Ay(b-y),\\ \sum_{n=1}^{\infty}\sin\frac{n\pi y}{b}\left(C_n\mathrm{e}^{\frac{n\pi a}{b}}+D_n\mathrm{e}^{-\frac{n\pi a}{b}}\right)&=0\end{aligned}$$

有

$$\begin{aligned}C_n+D_n&=\frac{2}{b}\int_0^b Ay(b-y)\sin\frac{n\pi y}{b}\mathrm{d}y\\ &=\frac{2A}{b}\left(\frac{b}{n\pi}\right)^2\int_0^b(b-2y)\mathrm{d}\left(\sin\frac{n\pi y}{b}\right)\\ &=\frac{4A}{b}\left(\frac{b}{n\pi}\right)^2\int_0^b\sin\frac{n\pi y}{b}\mathrm{d}y\\ &=\frac{4Ab^2}{n^3\pi^3}\left[1-(-1)^n\right] \\ C_n\mathrm{e}^{\frac{n\pi a}{b}}+D_n\mathrm{e}^{-\frac{n\pi a}{b}}&=0\end{aligned}$$

解得

$$\begin{cases}C_n=-\frac{2Ab^2}{n^3\pi^3}\left[1-(-1)^n\right]\frac{\mathrm{e}^{-\frac{n\pi a}{b}}}{\mathrm{sh}\frac{n\pi a}{b}}\\ D_n=\frac{2Ab^2}{n^3\pi^3}\left[1-(-1)^n\right]\frac{\mathrm{e}^{\frac{n\pi a}{b}}}{\mathrm{sh}\frac{n\pi a}{b}}\end{cases}$$

从而有

$$\begin{aligned}v(x,y)&=\sum_{n=0}^{\infty}\sin\frac{n\pi y}{b}\left\{-\frac{2Ab^2}{n^3\pi^3}\left[1-(-1)^n\right]\frac{\mathrm{e}^{-\frac{n\pi a}{b}}}{\mathrm{sh}\frac{n\pi a}{b}}\mathrm{e}^{\frac{n\pi x}{b}}+\frac{2Ab^2}{n^3\pi^3}\left[1-(-1)^n\right]\frac{\mathrm{e}^{\frac{n\pi a}{b}}}{\mathrm{sh}\frac{n\pi a}{b}}\mathrm{e}^{-\frac{n\pi x}{b}}\right\}\\ &=\sum_{n=1}^{\infty}\frac{4Ab^2}{n^3\pi^3}\left[1-(-1)^n\right]\sin\frac{n\pi y}{b}\frac{\mathrm{sh}\frac{n\pi(a-x)}{b}}{\mathrm{sh}\frac{n\pi a}{b}}\\ &=\sum_{k=0}^{\infty}\frac{8Ab^2}{(2k+1)^3\pi^3}\sin\frac{(2k+1)\pi y}{b}\frac{\mathrm{sh}\frac{(2k+1)\pi(a-x)}{b}}{\mathrm{sh}\frac{(2k+1)\pi a}{b}}\end{aligned}$$

综合两个解有

$$\begin{aligned}u(x,y)&=B\frac{\mathrm{sh}\frac{\pi(b-y)}{a}}{\mathrm{sh}\frac{\pi b}{a}}\sin\frac{\pi x}{a}\\ &\quad +\frac{8Ab^2}{\pi^3}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^3}\frac{\mathrm{sh}\frac{(2k+1)\pi(a-x)}{b}}{\mathrm{sh}\frac{(2k+1)\pi a}{b}}\sin\frac{(2k+1)\pi y}{b}。 \end{aligned}$$

12. 均匀的薄板占据区域 $0 < x < a, 0 < y < \infty$ 。边界上的温度:

$$u|_{x=0} = 0, u|_{x=a} = 0, u|_{y=0} = u_0, \lim_{y \rightarrow \infty} u = 0。$$

求解板的稳定温度分布。

解：定解问题为

$$\begin{aligned} u_{xx} + u_{yy} &= 0 \\ u_{x=0} &= 0, \quad u|_{x=a} = 0 \\ u|_{y=0} &= u_0, \quad u|_{y=\infty} = 0 \end{aligned}$$

类似教材本节例题 3，问题的一般解为

$$u(x, y) = \sum_{n=1}^{\infty} \sin \frac{n\pi x}{a} \left(C_n e^{\frac{n\pi y}{a}} + D_n e^{-\frac{n\pi y}{a}} \right)。$$

将此解代入 y 方向边值，由于要求 $\lim_{y \rightarrow \infty} u = 0$ ，而上述一般解中 $e^{\frac{n\pi y}{a}}$ 指数发散，因而置 $C_n = 0$ ，有

$$u(x, y) = \sum_{n=1}^{\infty} D_n e^{-\frac{n\pi y}{a}} \sin \frac{n\pi x}{a}$$

其中

$$\begin{aligned} D_n &= \frac{2}{a} \int_0^a u_0 \sin \frac{n\pi x}{a} dx \\ &= \frac{2u_0}{n\pi} \left[1 - (-1)^n \right] \\ &= \begin{cases} \frac{4u_0}{(2k+1)\pi}, & (n \text{ 为奇数 } 2k+1) \\ 0, & (n \text{ 为偶数}) \end{cases} \end{aligned}$$

从而，解为

$$u(x, y) = \frac{4u_0}{\pi} \sum_{k=0}^{\infty} \frac{1}{(2k+1)} e^{-\frac{(2k+1)\pi y}{a}} \sin \frac{(2k+1)\pi x}{a}。$$

13. 在带状区域 $0 < x < a, 0 < y < \infty$ 上求解 $\Delta u = 0$ ，并满足边界条件：

$$u|_{x=0} = 0, \quad u|_{x=a} = 0, \quad u|_{y=0} = A\left(1 - \frac{x}{a}\right), \quad \lim_{y \rightarrow \infty} u = 0。$$

解：定解问题为

$$\begin{aligned} u_{xx} + u_{yy} &= 0 \\ u_{x=0} &= 0, \quad u|_{x=a} = 0 \\ u|_{y=0} &= A\left(1 - \frac{x}{a}\right), \quad u|_{y=\infty} = 0 \end{aligned}$$

类似上题，问题的一般解为

$$u(x, y) = \sum_{n=1}^{\infty} \sin \frac{n\pi x}{a} \left(C_n e^{\frac{n\pi y}{a}} + D_n e^{-\frac{n\pi y}{a}} \right)。$$

将此解代入 y 方向边值，由于要求 $\lim_{y \rightarrow \infty} u = 0$ ，而上述定解中 $e^{\frac{n\pi y}{a}}$ 指数发散，因而置 $C_n = 0$ ，有

$$u(x, y) = \sum_{n=1}^{\infty} D_n e^{-\frac{n\pi y}{a}} \sin \frac{n\pi x}{a}$$

其中

$$\begin{aligned} D_n &= \frac{2}{a} \int_0^a A \left(1 - \frac{x}{a}\right) \sin \frac{n\pi x}{a} dx \\ &= \frac{2A}{a^2} \left(\frac{a}{n\pi}\right) \int_0^a (x-a) d\left(\cos \frac{n\pi x}{a}\right) \\ &= \frac{2A}{a^2} \left(\frac{a}{n\pi}\right) a \\ &= \frac{2A}{n\pi} \end{aligned}$$

从而，解为

$$u(x, y) = \frac{2A}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} e^{-\frac{n\pi y}{a}} \sin \frac{n\pi x}{a}。$$

14. 矩形膜的边长分别为 l_1 和 l_2 ，边缘固定。求它的本征振动模式。

解：定解问题为

$$u_{tt} - a^2(u_{xx} + u_{yy}) = 0, \tag{①}$$

$$u|_{x=0} = u|_{x=l_1} = 0, \tag{②}$$

$$u|_{y=0} = u|_{y=l_2} = 0。 \tag{③}$$

令 $u = T(t)w(x, y)$ ，代入定解问题

$$\begin{aligned} \frac{T''}{a^2 T} &= \frac{\Delta w}{w} = -\lambda \\ \begin{cases} \Delta w + \lambda w = 0 \\ T'' + \lambda a^2 T = 0 \end{cases} \end{aligned}$$

令 $w(x, y) = X(x)Y(y)$ ，分离变数

$$\begin{aligned} -\frac{X''}{X} &= \frac{Y''}{Y} + \lambda = \mu \\ \begin{cases} X'' + \mu X = 0 \\ Y'' + (\lambda - \mu)Y = 0 \end{cases} \end{aligned}$$

考虑边界条件 $X(0) = X(l_1) = Y(0) = Y(l_2) = 0$ ，有

$$\begin{aligned} \mu_m &= \frac{m^2 \pi^2}{l_1^2}, \quad \lambda_{nm} - \mu_m = \frac{n^2 \pi^2}{l_2^2} \\ \lambda_{nm} &= \frac{m^2 \pi^2}{l_1^2} + \frac{n^2 \pi^2}{l_2^2} \end{aligned}$$

因而本征函数为

$$\begin{aligned} X_m(x) &= \sin \frac{m\pi x}{l_1}, \quad Y_n(y) = \sin \frac{n\pi y}{l_2} \\ T_{nm} &= A_{nm} \cos \sqrt{\frac{m^2 \pi^2}{l_1^2} + \frac{n^2 \pi^2}{l_2^2}} at + B_{nm} \sin \sqrt{\frac{m^2 \pi^2}{l_1^2} + \frac{n^2 \pi^2}{l_2^2}} at \end{aligned}$$

所以，本征振动模式为

$$(A_{mn} \cos \omega_{mn} t + B_{mn} \sin \omega_{mn} t) \sin \frac{m\pi x}{l_1} \sin \frac{n\pi y}{l_2},$$

其中

$$\omega_{mn} = \pi a \sqrt{\left(\frac{m}{l_1}\right)^2 + \left(\frac{n}{l_2}\right)^2}。$$

除边缘外，平行于 y 轴的节线有 $m-1$ 条，即 $x = \frac{l_1}{m}$ 的整倍数；平行于 x 轴的节线有 $n-1$ 条，即 $y = \frac{l_2}{n}$ 的整倍数。节线保持不动，节线两侧振动方向相反。

15. 细圆环，半径为 ρ_0 ，初始温度分布已知为 $f(\varphi)$ ， φ 是以环心为极点的极角。环的表面是绝热的。求解环内温度变化的情况。
解：定解问题为

$$\begin{aligned} u_t - \left(\frac{a}{\rho_0}\right)^2 u_{\varphi\varphi} &= 0 \\ u|_{\varphi=0} &= u|_{\varphi=2\pi} \\ u|_{t=0} &= f(\varphi) \end{aligned}$$

考虑到 φ 的周期性条件，设解为

$$u(\varphi,t)=\sum_{n=0}^{\infty}T(t)(A_n\cos n\varphi+B_n\sin n\varphi)。$$

代入泛定方程，有

$$T'+\frac{n^2a^2}{\rho_0^2}T=0,$$

其解为

$$T_n(t)=C_n\mathrm{e}^{-\frac{n^2a^2t}{\rho^2}}。$$

这样，问题的一般解是

$$u(\varphi,t)=A_0+\sum_{n=1}^{\infty}(A_n\cos n\varphi+B_n\sin n\varphi)\mathrm{e}^{-\frac{n^2a^2t}{\rho^2}}。$$

代入初值，有

$$u|_{t=0}=A_0+\sum_{n=1}^{\infty}(A_n\cos n\varphi+B_n\sin n\varphi)=f(\varphi)，$$

解中各项系数为

$$\begin{cases} A_0=\frac{1}{2\pi}\int_0^{2\pi}f(\varphi)\mathrm{d}\varphi \\ A_n=\frac{1}{\pi}\int_0^{2\pi}f(\varphi)\cos n\varphi\mathrm{d}\varphi \\ B_n=\frac{1}{\pi}\int_0^{2\pi}f(\varphi)\sin n\varphi\mathrm{d}\varphi \end{cases}$$

16. A 、 B 为常数，在半径为 ρ_0 的圆形域内求解 $\Delta u = 0$ ，并分别满足边界条件：

$$(1) u|_{\rho=\rho_0} = A|\sin\varphi|, \quad (2) u|_{\rho=\rho_0} = A + B\sin\varphi.$$

解：类似教材本节例 4，圆形区域内，拉普拉斯方程的一般解为

$$u(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{n=1}^{\infty} (C_n \rho^n + D_n \rho^{-n}) (A_n \cos n\varphi + B_n \sin n\varphi).$$

要使得 $\rho \rightarrow 0$ 有限，必须置 $D_n = 0$ ，因而

$$u(\rho, \varphi) = C_0 + \sum_{n=1}^{\infty} \rho^n (A_n \cos n\varphi + B_n \sin n\varphi). \tag{①}$$

(1) 将①代入边界条件有

$$C_0 + \sum_{n=1}^{\infty} \rho_0^n (A_n \cos n\varphi + B_n \sin n\varphi) = A|\sin\varphi|.$$

$$C_0 = \frac{A}{\pi} \int_0^\pi \sin\varphi d\varphi = \frac{A}{\pi} (-\cos\varphi) \Big|_0^\pi = \frac{2A}{\pi},$$

$$\begin{aligned} A_n &= \frac{2A}{\rho_0^n \pi} \int_0^\pi \sin\varphi \cos n\varphi d\varphi \\ &= \frac{2A}{\rho_0^n \pi} \left[-\frac{\cos(1-n)\varphi}{2(1-n)} - \frac{\cos(1+n)\varphi}{2(1+n)} \right] \Big|_0^\pi = \begin{cases} \frac{2A}{\rho_0^{2k} \pi}, & (n=2k) \\ 0, & (n=2k+1) \end{cases} \end{aligned}$$

$$B_n = 0.$$

解是

$$u(\rho, \varphi) = \frac{2A}{\pi} + \frac{4A}{\pi} \sum_{k=1}^{\infty} \frac{1}{1-4k^2} \left(\frac{\rho}{\rho_0} \right)^{2k} \cos 2k\varphi.$$

(2) 将①代入边界条件有

$$u|_{\rho=\rho_0} = C_0 + \sum_{n=1}^{\infty} \rho_0^n (A_n \cos n\varphi + B_n \sin n\varphi) = A + B\sin\varphi,$$

根据傅里叶级数的正交关系，有

$$C_0 = A, \quad \rho_0 B_1 = B,$$

其余各项系数均为零。所以，解为

$$u(\rho, \varphi) = A + B \frac{\rho}{\rho_0} \sin\varphi.$$

17. 半圆形薄板，半径为 ρ_0 ，板面绝热，边界直径上温度保持零度，圆周上保持 u_0 。求稳定状态下的板上温度分布。

解：定解问题是

$$\begin{aligned} \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} &= 0, & (\rho \in (0, \rho_0), 0 \leq \varphi \leq \pi) \\ u|_{\rho=\rho_0} &= u_0, \\ u|_{\varphi=0, \pi} &= 0. \end{aligned}$$

令 $u = R(\rho)\Phi(\varphi)$ ，代入定解问题，分离变数，得

$$\begin{cases} \Phi'' + \lambda \Phi = 0, \\ \Phi(0) = \Phi(\pi) = 0. \end{cases} \tag{①}$$

$$\rho^2 R'' + \rho R' - \lambda R = 0. \tag{②}$$

本征值问题①的本征函数是 $\Phi_n = \sin n\varphi$ ，本征值是 $\lambda = n^2$ ($n=1,2,\dots$)。方程②的解是

$$R_n(\rho) = C_n \rho^n + D_n \rho^{-n},$$

从而问题的一般解为

$$u(\rho, \varphi) = \sum_{n=1}^\infty (C_n \rho^n + D_n \rho^{-n}) \sin n\varphi$$

代入边值定系数，由圆内有界条件，置 $D_n=0$,

$$B_n \rho_0^n = \frac{2}{\pi} \int_0^\pi u_0 \sin n\varphi \, d\varphi = \frac{2u_0}{n\pi} \left[1 - (-1)^n \right].$$

从而，解为

$$u(\rho, \varphi) = \frac{4u_0}{\pi} \sum_{k=0}^\infty \frac{1}{2k+1} \left(\frac{\rho}{\rho_0} \right)^{2k+1} \sin(2k+1)\varphi.$$

18. 把例 4 的导体圆柱换为介质圆柱, 介质的介电常数为 ε 。求解柱内外的电场。(提示: 柱内电势必须有限, 在柱面上电势连续, 电位移的法向分量连续)

解: 坐标系统建立同例 4, 设柱内电势为 u^I , 柱外电势为 u^{II} , 定解问题为

$$\begin{cases} \frac{\partial^2 u^I}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u^I}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u^I}{\partial \varphi^2} = 0, \\ u^I|_{\rho \rightarrow 0} \text{ 有界} \end{cases} \quad \begin{cases} \frac{\partial^2 u^{II}}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u^{II}}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u^{II}}{\partial \varphi^2} = 0, \\ u^{II}|_{\rho \rightarrow \infty} = -E_0 \rho \cos \varphi + \frac{q_0}{2\pi\varepsilon_0} \ln \frac{1}{\rho} + u_0 \end{cases}$$

$$u^I|_{\rho=\rho_0} = u^{II}|_{\rho=\rho_0}, \quad \varepsilon \frac{\partial u^I}{\partial \rho} \bigg|_{\rho=\rho_0} = \varepsilon_0 \frac{\partial u^{II}}{\partial \rho} \bigg|_{\rho=\rho_0} \quad (\text{此处将柱体半径记为 } \rho_0)$$

其中 $u^{II}|_{\rho \rightarrow \infty} = \frac{q_0}{2\pi\varepsilon_0} \ln \frac{1}{\rho}$ 是考虑柱体单位长度带电荷量 q_0 在周围空间产生的电势。 u_0 为常数,

由电势零点的选取决定。柱内外电势的一般解为

$$u^I(\rho, \varphi) = c_0 + d_0 \ln \rho + \sum_{n=1}^{\infty} (c_n \rho^n + d_n \rho^{-n}) (a_n \cos n\varphi + b_n \sin n\varphi),$$

$$u^{II}(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{n=1}^{\infty} (C_n \rho^n + D_n \rho^{-n}) (A_n \cos n\varphi + B_n \sin n\varphi)。$$

考虑到圆柱内, 在 $\rho \rightarrow 0$ 内有限, 柱内电势一般解为

$$u^I(\rho, \varphi) = c_0 + \sum_{n=1}^{\infty} \rho^n (a_n \cos n\varphi + b_n \sin n\varphi)。$$

考虑在圆柱外 $u^{II}|_{\rho \rightarrow \infty} = -E_0 \rho \cos \varphi + \frac{q_0}{2\pi\varepsilon_0} \ln \frac{1}{\rho} + u_0$, 其一般解

$$u^{II}(\rho, \varphi) \big|_{\rho \rightarrow \infty} = C_0 + D_0 \ln \rho + \sum_{n=1}^{\infty} C_n \rho^n (A_n \cos n\varphi + B_n \sin n\varphi)。$$

上两式右端比较可知,

$$C_0 = u_0, \quad D_0 = -\frac{q_0}{2\pi\varepsilon_0}, \quad C_1 A_1 = -E_0, \quad C_n A_n = 0 \quad (n \neq 1), \quad C_n B_n = 0。$$

从而 $u^{II}(\rho, \varphi) = u_0 - \frac{q_0}{2\pi\varepsilon_0} \ln \rho - E_0 \rho \cos \varphi + \sum_{n=1}^{\infty} \rho^{-n} (A_n \cos n\varphi + B_n \sin n\varphi)$, 代入衔接条件,

有

$$\left\{ \begin{aligned} c_0 + \sum_{n=1}^{\infty} \rho_0^n (a_n \cos n\varphi + b_n \sin n\varphi) &= u_0 - \frac{q_0}{2\pi\varepsilon_0} \ln \rho_0 + (-E_0\rho_0 + A_1\rho_0^{-1}) \cos \varphi \\ &\quad + B_1\rho_0^{-1} \sin \varphi + \sum_{n=2}^{\infty} \rho_0^{-n} (A_n \cos n\varphi + B_n \sin n\varphi) \\ \varepsilon \sum_{n=1}^{\infty} n \rho_0^{n-1} (a_n \cos n\varphi + b_n \sin n\varphi) &= -\frac{q_0}{2\pi\rho_0} + (-E_0 - A_1\rho_0^{-2}) \cos \varphi \\ &\quad - B_1\rho_0^{-2} \sin \varphi + \sum_{n=2}^{\infty} (-n) \rho_0^{-n-1} (A_n \cos n\varphi + B_n \sin n\varphi) \end{aligned} \right.$$

比较两边系数，得

$$\begin{aligned} c_0 &= u_0 - \frac{q_0}{2\pi\varepsilon_0} \ln \rho_0 \\ \begin{cases} a_1\rho_0 = -E_0\rho_0 + A_1\rho_0^{-1} \\ \varepsilon a_1 = -E_0 - \rho_0^{-2} A_1 \end{cases} \\ \begin{cases} b_1\rho_0 = B_1\rho_0^{-1} \\ \varepsilon b_1 = -\rho_0^{-2} B_1 \end{cases} \\ \begin{cases} a_n\rho_0^n = A_n\rho_0^{-n} \\ \varepsilon n\rho_0^{n-1} a_n = -n\rho_0^{-n-1} A_n \end{cases} \\ \begin{cases} b_n\rho_0^n = B_n\rho_0^{-n} \\ \varepsilon n\rho_0^{n-1} b_n = -n\rho_0^{-n-1} B_n \end{cases} \end{aligned}$$

解得

$$\begin{cases} a_1 = -\frac{2E_0}{\varepsilon+1} \\ A_1 = \frac{\varepsilon-1}{\varepsilon+1} \rho_0^2 E_0 \end{cases}$$

其余均为零。最后的电势的解为

$$\begin{aligned} u^{\text{I}}(\rho, \varphi) &= u_0 - \frac{q_0}{2\pi\varepsilon_0} \ln \rho_0 - \frac{2E_0}{\varepsilon+1} \rho \cos \varphi, \\ u^{\text{II}}(\rho, \varphi) &= u_0 - \frac{q_0}{2\pi\varepsilon_0} \ln \rho_0 - \left(\rho - \frac{\varepsilon-1}{\varepsilon+1} \frac{\rho_0^2}{\rho} \right) E_0 \sin \varphi \end{aligned}$$

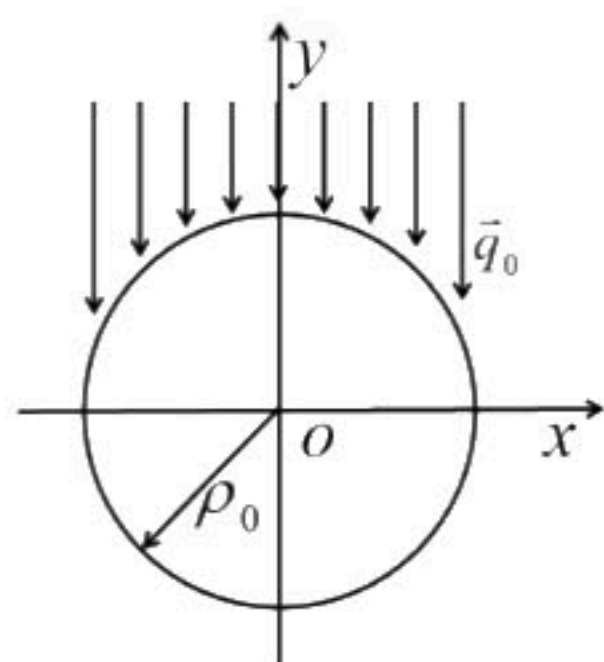
若柱面不带自由电荷，且选柱面为电势零点，则解为

$$\begin{aligned} u^{\text{I}}(\rho, \varphi) &= u_{\text{内}}(\rho, \varphi) = -\frac{2E_0}{\varepsilon+1} \rho \cos \varphi, \\ u^{\text{II}}(\rho, \varphi) &= u_{\text{外}}(\rho, \varphi) = -\left(\rho - \frac{\varepsilon-1}{\varepsilon+1} \frac{\rho_0^2}{\rho} \right) E_0 \sin \varphi \end{aligned}$$

柱内为匀强电场， $E = \frac{2}{1+\varepsilon} E_0$ 。柱内极化强度 $P = (\varepsilon-1)\varepsilon_0 E = 2\varepsilon_0 \frac{\varepsilon-1}{\varepsilon+1} E_0$ 。柱面束缚电荷面密度
 $= P$ 的法向分量 $= 2\varepsilon_0 \frac{\varepsilon-1}{\varepsilon+1} E_0 \cos \varphi$ 。

19. 半径为 ρ_0 ，表面熏黑了的均匀长圆柱，在温度为零度的空气中受着阳光照射. 阳光垂直于柱轴，热流强度为 q_0 （图 8-7）。试求柱内稳定温度分布。（提示：泛定方程为 $\Delta u = 0$ ，边界条件为 $(ku_\rho + hu)|_{\rho=\rho_0} = f(\varphi)$ $f(\varphi)$ 是热流强度的法向分量。）如取极轴垂直于阳光，则

$$f(\varphi) = \begin{cases} q_0 \sin \varphi, & (0 < \varphi < \pi) \\ 0, & (\pi < \varphi < 2\pi) \end{cases}$$



解：定解问题为

$$\Delta u = 0,$$

$$(ku_\rho + hu)|_{\rho=\rho_0} = f(\varphi) = \begin{cases} q_0 \sin \varphi, & (0 < \varphi < \pi) \\ 0, & (\pi < \varphi < 2\pi) \end{cases}$$

本问题圆柱内的有界解为

$$u(\rho, \varphi) = C_0 + \sum_{n=1}^{\infty} \rho^n (A_n \cos n\varphi + B_n \sin n\varphi)$$

代入边界条件后有

$$\begin{aligned} (ku_\rho + hu)|_{\rho=\rho_0} &= k \sum_{n=1}^{\infty} n \rho_0^{n-1} (A_n \cos n\varphi + B_n \sin n\varphi) \\ &\quad + hC_0 + \sum_{n=1}^{\infty} h \rho_0^n (A_n \cos n\varphi + B_n \sin n\varphi) \\ &= hC_0 + \sum_{n=1}^{\infty} (kn \rho_0^{n-1} + h \rho_0^n) (A_n \cos n\varphi + B_n \sin n\varphi) \end{aligned}$$

因而

$$\begin{cases} (kn \rho_0^{n-1} + h \rho_0^n) A_n = \frac{q_0}{\pi} \int_0^\pi \sin \varphi \cos n\varphi d\varphi = \left[(-1)^{n-1} - 1 \right] \frac{q_0}{\pi(n^2 - 1)} \\ (kn \rho_0^{n-1} + h \rho_0^n) B_n = \frac{q_0}{\pi} \int_0^\pi \sin \varphi \sin n\varphi d\varphi = \frac{\pi q_0}{2} \delta_{n1} \\ hC_0 = \frac{1}{2\pi} \int_0^\pi q_0 \sin \varphi d\varphi = \frac{q_0}{\pi} \end{cases}$$

解得

$$\begin{cases} A_n = \begin{cases} 0, & (n = 2m+1) \\ \frac{2q_0}{\pi(1-4m^2)(2km\rho_0^{2m-1} + h\rho_0^{2m})}, & (n = 2m) \end{cases} \\ B_1 = \frac{q_0}{2(k + h\rho_0)}, \quad B_{n \neq 1} = 0 \\ C_0 = \frac{q_0}{\pi h} \end{cases}$$

最终，解为

$$\begin{aligned} u(\rho, \varphi) &= \frac{q_0}{\pi h} + \frac{1}{k + h\rho_0} \frac{q_0}{2} \rho \sin \varphi \\ &\quad + \sum_{m=1}^{\infty} \frac{2q_0}{\pi \rho_0^{2m-1} (1-4m^2)(2mk + h\rho_0)} \rho^{2m} \cos 2m\varphi. \end{aligned}$$

解得

$$\begin{cases} A_n = \begin{cases} 0, & (n = 2m+1) \\ \frac{2q_0}{\pi(1-4m^2)(2km\rho_0^{2m-1} + h\rho_0^{2m})}, & (n = 2m) \end{cases} \\ B_1 = \frac{q_0}{2(k+h\rho_0)}, \quad B_{n\neq 1} = 0 \\ C_0 = \frac{q_0}{\pi h} \end{cases}$$

最终，解为

$$\begin{aligned} u(\rho, \varphi) = & \frac{q_0}{\pi h} + \frac{1}{k+h\rho_0} \frac{q_0}{2} \rho \sin \varphi \\ & + \sum_{m=1}^{\infty} \frac{2q_0}{\pi \rho_0^{2m-1} (1-4m^2)(2mk+h\rho_0)} \rho^{2m} \cos 2m\varphi. \end{aligned}$$

20. 在以原点为中心, 以 ρ_1 和 ρ_2 为半径的两个同心圆围成的环域上求解 $\Delta u = 0$, 使其满足边界条件 $u|_{\rho=\rho_1} = f_1(\varphi), u|_{\rho=\rho_2} = f_2(\varphi)$ 。

解: 取极坐标, 定解问题为

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} = 0, \quad (\rho_1 < \rho < \rho_2)$$

$$u|_{\rho=\rho_1} = f_1(\varphi), \quad u|_{\rho=\rho_2} = f_2(\varphi)。$$

本问题的一般解为

$$u(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{n=1}^{\infty} (C_n \rho^n + D_n \rho^{-n}) (A_n \cos n\varphi + B_n \sin n\varphi)。$$

代入边界条件有

$$\begin{cases} C_0 + D_0 \ln \rho_1 + \sum_{n=1}^{\infty} (C_n \rho_1^n + D_n \rho_1^{-n}) (A_n \cos n\varphi + B_n \sin n\varphi) = f_1(\varphi) \\ C_0 + D_0 \ln \rho_2 + \sum_{n=1}^{\infty} (C_n \rho_2^n + D_n \rho_2^{-n}) (A_n \cos n\varphi + B_n \sin n\varphi) = f_2(\varphi) \end{cases}$$

将两个边界条件展开为傅里叶级数, 有

$$\begin{cases} C_0 + D_0 \ln \rho_1 = \frac{1}{2\pi} \int_0^{2\pi} f_1(\varphi) d\varphi \equiv \beta_0^{(1)} \\ C_0 + D_0 \ln \rho_2 = \frac{1}{2\pi} \int_0^{2\pi} f_2(\varphi) d\varphi \equiv \beta_0^{(2)} \\ (C_n \rho_1^n + D_n \rho_1^{-n}) A_n = \frac{1}{2\pi} \int_0^{2\pi} f_1(\varphi) \cos n\varphi d\varphi \equiv \beta_n^{(1)} \\ (C_n \rho_2^n + D_n \rho_2^{-n}) A_n = \frac{1}{2\pi} \int_0^{2\pi} f_2(\varphi) \cos n\varphi d\varphi \equiv \beta_n^{(2)} \\ (C_n \rho_1^n + D_n \rho_1^{-n}) B_n = \frac{1}{2\pi} \int_0^{2\pi} f_1(\varphi) \sin n\varphi d\varphi \equiv \alpha_n^{(1)} \\ (C_n \rho_2^n + D_n \rho_2^{-n}) B_n = \frac{1}{2\pi} \int_0^{2\pi} f_2(\varphi) \sin n\varphi d\varphi \equiv \alpha_n^{(2)} \end{cases}$$

$$\begin{cases} C_0 + D_0 \ln \rho_1 = \beta_0^{(1)} \\ C_0 + D_0 \ln \rho_2 = \beta_0^{(2)} \end{cases} \Rightarrow \begin{cases} C_0 = \frac{\beta_0^{(1)} \ln \rho_2 - \beta_0^{(2)} \ln \rho_1}{\ln \rho_2 - \ln \rho_1} \\ D_0 = \frac{\beta_0^{(1)} - \beta_0^{(2)}}{\ln \rho_1 - \ln \rho_2} \end{cases}$$

$$\begin{cases} (C_n \rho_1^n + D_n \rho_1^{-n}) A_n = \beta_n^{(1)} \\ (C_n \rho_2^n + D_n \rho_2^{-n}) A_n = \beta_n^{(2)} \end{cases} \Rightarrow \begin{cases} A_n C_n = \frac{\rho_2^{-n} \beta_n^{(1)} - \rho_1^{-n} \beta_n^{(2)}}{\rho_1^n \rho_2^{-n} - \rho_1^{-n} \rho_2^n} \\ A_n D_n = -\frac{\rho_2^n \beta_n^{(1)} - \rho_1^n \beta_n^{(2)}}{\rho_1^n \rho_2^{-n} - \rho_1^{-n} \rho_2^n} \end{cases}$$

$$\begin{cases} (C_n \rho_1^n + D_n \rho_1^{-n}) B_n = \alpha_n^{(1)} \\ (C_n \rho_2^n + D_n \rho_2^{-n}) B_n = \alpha_n^{(2)} \end{cases} \Rightarrow \begin{cases} B_n C_n = \frac{\rho_2^{-n} \alpha_n^{(1)} - \rho_1^{-n} \alpha_n^{(2)}}{\rho_1^n \rho_2^{-n} - \rho_1^{-n} \rho_2^n} \\ B_n D_n = -\frac{\rho_1^{-n} \alpha_n^{(1)} - \rho_2^{-n} \alpha_n^{(2)}}{\rho_1^n \rho_2^{-n} - \rho_1^{-n} \rho_2^n} \end{cases}$$

从而，问题的解为

$$\begin{aligned} u(\rho, \varphi) = & \frac{\beta_0^{(1)} \ln \rho_2 - \beta_0^{(2)} \ln \rho_1}{2(\ln \rho_2 - \ln \rho_1)} + \frac{\beta_0^{(2)} - \beta_0^{(1)}}{2(\ln \rho_2 - \ln \rho_1)} \ln \rho \\ & + \sum_{n=1}^{\infty} \left[\frac{\beta_n^{(1)} \rho_2^{-n} - \beta_n^{(2)} \rho_1^{-n}}{\rho_1^n \rho_2^{-n} - \rho_1^{-n} \rho_2^n} \rho^n - \frac{\beta_n^{(1)} \rho_2^n - \beta_n^{(2)} \rho_1^n}{\rho_1^n \rho_2^{-n} - \rho_1^{-n} \rho_2^n} \rho^{-n} \right] \cos n\varphi \\ & + \sum_{n=1}^{\infty} \left[\frac{\alpha_n^{(1)} \rho_2^{-n} - \alpha_n^{(2)} \rho_1^{-n}}{\rho_1^n \rho_2^{-n} - \rho_1^{-n} \rho_2^n} \rho^n - \frac{\alpha_n^{(1)} \rho_2^n - \alpha_n^{(2)} \rho_1^n}{\rho_1^n \rho_2^{-n} - \rho_1^{-n} \rho_2^n} \rho^{-n} \right] \sin n\varphi. \end{aligned}$$

$$\text{其中 } \beta_0^{(1)} = \frac{1}{\pi} \int_{-\pi}^{\pi} f_1(\varphi) d\varphi, \quad \beta_0^{(2)} = \frac{1}{\pi} \int_{-\pi}^{\pi} f_2(\varphi) d\varphi,$$

$$\beta_n^{(1)} = \frac{1}{\pi} \int_{-\pi}^{\pi} f_1(\varphi) \cos n\varphi d\varphi, \quad \beta_n^{(2)} = \frac{1}{\pi} \int_{-\pi}^{\pi} f_2(\varphi) \cos n\varphi d\varphi,$$

$$\alpha_n^{(1)} = \frac{1}{\pi} \int_{-\pi}^{\pi} f_1(\varphi) \sin n\varphi d\varphi, \quad \alpha_n^{(2)} = \frac{1}{\pi} \int_{-\pi}^{\pi} f_2(\varphi) \sin n\varphi d\varphi.$$

21. 求解绕圆柱的水流问题。在远离圆柱因而未受圆柱干扰处的水流是均匀的，流速为 v_0 。圆柱半径为 a 。

解：水无旋，速度势满足拉普拉斯方程，取极坐标，定解问题为

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} = 0, \quad (\rho_1 < \rho < \rho_2)$$

$$\left. \frac{\partial u}{\partial \rho} \right|_{\rho=a} = 0, \quad u|_{\rho \rightarrow \infty} = v_0 \cos \varphi.$$

$$u(\rho, \varphi) = C_0 + D_0 \ln \rho + A_0 \varphi + \sum_{n=1}^{\infty} (C_n \rho^n + D_n \rho^{-n}) (A_n \cos n\varphi + B_n \sin n\varphi).$$

代入 $u|_{\rho=a} = 0$ 后有

$$\begin{cases} D_0 = 0 \\ C_n a^{2n} - D_n = 0 \end{cases}$$

令常数 $C_0 = 0$ ，解简化为

$$u(\rho, \varphi) = A_0 \varphi + \sum_{n=1}^{\infty} (\rho^n + a^{2n} \rho^{-n}) (A_n \cos n\varphi + B_n \sin n\varphi).$$

考虑 $u|_{\rho \rightarrow \infty} = v_0 \rho \cos \varphi$ ，有

$$A_1 = v_0, A_{n \neq 1} = 0, B_n = 0.$$

记 $A_0 = \frac{\Gamma}{2\pi}$ ， Γ 任意，则速度势为

$$u(\rho, \varphi) = v_0 \rho \cos \varphi + v_0 a^2 \rho^{-1} \cos \varphi + \frac{\Gamma}{2\pi} \varphi.$$

22. 长为 l 的理想传输线，一端接于电动势为 $v_0 \sin \omega t$ 的交流电源，另一端短路。求解线上的稳恒电振荡并计算输入阻抗。

解：本题为无初值问题，定解问题为

$$\begin{aligned} v_{tt} - a^2 v_{xx} &= 0, \quad a = \frac{1}{\sqrt{LC}} \\ v|_{x=0} &= v_0 e^{i\omega t}, \quad v|_{x=l} = 0. \end{aligned}$$

将边值 $v_0 \sin \omega t$ 写成 $v_0 e^{i\omega t}$ ，需在结果中取实部。这是典型的受迫振动，我们考虑传输线上有同步的稳恒电振荡，设解为

$$v = X(x) e^{i\omega t}。$$

代入原方程后有

$$-\omega^2 X(x) e^{i\omega t} - a^2 X''(x) e^{i\omega t} = 0。$$

化简得

$$\begin{aligned} X''(x) + \frac{\omega^2}{a^2} X(x) &= 0, \\ X(x) &= A e^{i \frac{\omega x}{a}} + B e^{-i \frac{\omega x}{a}}。 \end{aligned}$$

从而

$$v(x, t) = \left(A e^{i \frac{\omega x}{a}} + B e^{-i \frac{\omega x}{a}} \right) e^{i\omega t}。$$

代入边界条件

$$\begin{cases} A + B = v_0 \\ A e^{i \frac{\omega l}{a}} + B e^{-i \frac{\omega l}{a}} = 0 \end{cases}$$

解得

$$\begin{cases} A = \frac{i v_0 e^{-i \frac{\omega l}{a}}}{2 \sin \frac{\omega l}{a}} \\ B = \frac{-i v_0 e^{i \frac{\omega l}{a}}}{2 \sin \frac{\omega l}{a}} \end{cases}$$

因而，稳定的电振荡为

$$\begin{aligned}
 v(x,t) &= \frac{e^{\frac{j\omega}{a}(x-l)} - e^{-\frac{j\omega}{a}(x-l)}}{-2j \sin \frac{\omega l}{a}} v_0 e^{j\omega t} \\
 &= \frac{\sin \frac{\omega}{a}(l-x)}{\sin \frac{\omega l}{a}} v_0 e^{j\omega t} = \frac{\sin \omega \sqrt{LC}(l-x)}{\sin \omega l \sqrt{LC}} v_0 e^{j\omega t}。
 \end{aligned}$$

取虚部，有

$$v(x,t) = \frac{\sin \omega \sqrt{LC}(l-x)}{\sin \omega l \sqrt{LC}} v_0 \sin \omega t。$$

利用 $j_t = -\frac{1}{L} v_x$ ，

$$\begin{aligned}
 j_t &= -\frac{1}{L} v_x = \frac{\omega v_0 \cos \omega \sqrt{LC}(l-x)}{aL \sin \omega l \sqrt{LC}} \sin \omega t \\
 j &= \int j_t dt = -v_0 \sqrt{\frac{C}{L}} \frac{\cos \omega \sqrt{LC}(l-x)}{\sin \omega l \sqrt{LC}} \cos \omega t + f(t)
 \end{aligned}$$

上式对 t 求导后为 $-\frac{1}{L} v_x$ ，可知 $f'(t) = 0$ ，取 $f(t) = 0$ ，因而，输入阻抗为

$$Z_\lambda = \frac{v_{\max}|_{x=0}}{j_{\max}|_{x=0}} = \sqrt{\frac{L}{C}} \tan \omega l \sqrt{LC}。$$

若 $l = \frac{\lambda}{4}$ ， $\omega = \frac{2\pi a}{\lambda}$ ，则

$$Z_\lambda = \sqrt{\frac{L}{C}} \tan \left(\frac{2\pi a}{\lambda} \cdot \frac{\lambda}{4} \cdot \frac{1}{a} \right) = \sqrt{\frac{L}{C}} \tan \frac{\pi}{2} = \infty$$

23. 长为 l 的非理想传输线，一端接于交流电源 $v_0 \sin \omega t$ ，另一端通过阻抗元件 R_0 ， L_0 和 C_0 而相接。求解线上的稳恒电振荡。在怎样的条件下不存在反射波（这叫做匹配）？

解：传输线中电压满足的定解问题为

$$\begin{aligned} v_{tt} - a^2 v_{xx} + \frac{LG + RC}{LC} v_t + \frac{RG}{LC} v &= 0, \\ v|_{x=0} &= v_0 e^{i\omega t}, \\ v|_{x=l} &= \left(R_0 j + L_0 \frac{dj}{dt} + \frac{1}{C_0} \int_{t_0}^t j dt \right) \Big|_{x=l}. \end{aligned}$$

将边值 $v_0 \sin \omega t$ 写成 $v_0 e^{i\omega t}$ ，在结果中取实部。本问题在传输线上存在稳定的电振荡，设其解的形式为

$$v(x, t) = X(x) e^{i\omega t},$$

代入上述运动方程有

$$\begin{aligned} -\omega^2 X(x) e^{i\omega t} - a^2 X''(x) e^{i\omega t} \\ + i\omega \frac{LG + RC}{LC} X(x) e^{i\omega t} + \frac{RG}{LC} X(x) e^{i\omega t} = 0 \end{aligned}$$

整理得

$$X''(x) + \omega^2 LC X(x) - i\omega(LG + RC) X(x) - RG X(x) = 0$$

利用试探解 $X(x) = e^{\lambda x}$ 代入后

$$\lambda^2 = (R + i\omega L)(G + i\omega C) \Rightarrow \lambda = \sqrt{(R + i\omega L)(G + i\omega C)}$$

因而，

$$\begin{aligned} X(x) &= A e^{\lambda x} + B e^{-\lambda x} \\ v(x, t) &= X(x) e^{i\omega t} = (A e^{\lambda x} + B e^{-\lambda x}) e^{i\omega t} \end{aligned}$$

代入电流公式

$$j_x = -Gv - C v_t = -(G + i\omega C)(A e^{\lambda x} + B e^{-\lambda x}) e^{i\omega t}$$

积分后得

$$j = -\frac{G + i\omega C}{\lambda} (A e^{\lambda x} - B e^{-\lambda x}) e^{i\omega t} + f(t)$$

由 (7.1.10) $v_x = -Rj - Lj_t$ ，整理后，有

$$Lf'(t) + Rf(t) = 0 \Rightarrow f(t) = f_0 e^{-\frac{R}{L}t}.$$

考虑边界条件

$$\begin{cases} v|_{x=0} = v_0 e^{i\omega t}, \\ v|_{x=l} = j|_{x=l} R_0 + L_0 j_t|_{x=l} + \frac{1}{C_0} \int_{t_0}^t j|_{x=l} dt \end{cases}$$

有

$$\begin{cases} A + B = v_0 \\ Ae^{\lambda l} + Be^{-\lambda l} = -\frac{R_0 + i\omega L_0 + \frac{1}{i\omega C_0}}{\lambda} (G + i\omega C) (Ae^{\lambda l} - Be^{-\lambda l}) \\ f_0 = 0 \end{cases}$$

解得

$$\begin{cases} A = \frac{R_0 + i\omega L_0 + \frac{1}{i\omega C_0} - \sqrt{\frac{R + i\omega L}{G + i\omega C}}}{K} \\ B = \frac{R_0 + i\omega L_0 + \frac{1}{i\omega C_0} + \sqrt{\frac{R + i\omega L}{G + i\omega C}}}{K} \end{cases}$$

其中

$$K = \left(R_0 + i\omega L_0 + \frac{1}{i\omega C_0} + \sqrt{\frac{R + i\omega L}{G + i\omega C}} \right) e^{\lambda l} + \left(R_0 + i\omega L_0 + \frac{1}{i\omega C_0} - \sqrt{\frac{R + i\omega L}{G + i\omega C}} \right) e^{-\lambda l}。$$

注意到，解 $v(x, t) = X(x)e^{i\omega t} = Ae^{i\omega t + \lambda x} + Be^{i\omega t - \lambda x}$ 中，第一项代表反射项，向左的行波，而第二项为向右的行波是入射项。要使得传输线中没有反射项，必须有 $A = 0$ 。因而，没有反射波的条件为

$$R_0 + i\omega L_0 + \frac{1}{i\omega C_0} = \sqrt{\frac{R + i\omega L}{G + i\omega C}},$$

等号右边叫做传输线的特征阻抗 Z_0 。

24. 长为 l 的均匀杆，一端固定，另一端在纵向力 $F(t) = F_0 \sin \omega t$ 长期作用下。求解杆的稳恒纵振动。

解：这是无初值问题，杆的定解问题为

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= 0 \\ u|_{x=0} &= 0, u_x|_{x=l} = \frac{F_0}{ES} \sin \omega t \end{aligned}$$

外加力使杆做与其同步的强迫振动，设解为

$$u(x, t) = X(x) \sin \omega t$$

代入运动方程后有

$$X'' + \frac{\omega^2}{a^2} X = 0,$$

其解是

$$X(x) = A \cos \frac{\omega x}{a} + B \sin \frac{\omega x}{a}。$$

因而

$$u(x, t) = \left(A \cos \frac{\omega x}{a} + B \sin \frac{\omega x}{a} \right) \sin \omega t。$$

由边界条件有

$$\begin{aligned} u|_{x=0} &= 0 \Rightarrow A = 0 \\ u_x|_{x=l} &= \frac{F_0}{ES} \sin \omega t \Rightarrow B = \frac{aF_0}{\omega ES} \frac{1}{\cos \frac{\omega l}{a}} \end{aligned}$$

所以，稳定的振动解为

$$u(x, t) = \frac{aF_0}{\omega ES \cos(\omega l / a)} \sin \frac{\omega x}{a} \sin \omega t。$$

25.长为 l 的均匀弦两端固定, 在 x_0 点有一集中的横向力 $F(t) = A\cos\omega t$ 一直作用着, 求解弦的稳恒横振动。

解: 设作用点左端的位移为 u_1 , 右端的位移为 u_2 , 定解问题为

$$\begin{aligned} u_{1tt} - a^2 u_{1xx} &= 0, & u_{2tt} - a^2 u_{2xx} &= 0, \\ u_1|_{x=0} &= 0, & u_2|_{x=l} &= 0, \\ u_1|_{x=x_0} &= u_2|_{x=x_0}, \\ T(u_{2x} - u_{1x})|_{x=x_0} &= -A\cos\omega t. \end{aligned}$$

根据上一题, 弦上有稳定振动, 设

$$\begin{cases} u_1(x, t) = X_1(x)\cos\omega t \\ u_2(x, t) = X_2(x)\cos\omega t \end{cases}$$

代入定解问题

$$\begin{aligned} X_1'' + \frac{\omega^2}{a^2} X_1 &= 0, & X_2'' + \frac{\omega^2}{a^2} X_2 &= 0, \\ X_1|_{x=0} &= 0, & X_2|_{x=l} &= 0, \\ X_1|_{x=x_0} &= X_2|_{x=x_0}, \\ T(X_2' - X_1')|_{x=x_0} &= -A. \end{aligned}$$

$$X_1(x) = A_1 \cos \frac{\omega x}{a} + B_1 \sin \frac{\omega x}{a}, \quad X_2(x) = A_2 \cos \frac{\omega x}{a} + B_2 \sin \frac{\omega x}{a},$$

代入边界条件, 有

$$\begin{cases} A_1 = 0 \\ A_2 \cos \frac{\omega l}{a} + B_2 \sin \frac{\omega l}{a} = 0 \end{cases}$$

因而,

$$\begin{cases} X_1(x) = B_1 \sin \frac{\omega x}{a} \\ X_2(x) = A_2 \frac{\sin[\omega(l-x)/a]}{\sin[\omega l/a]} \end{cases}$$

代入衔接条件后有

$$\begin{cases} B_1 \sin \frac{\omega x_0}{a} = A_2 \frac{\sin[\omega(l-x_0)/a]}{\sin[\omega l/a]} \\ \frac{T\omega}{a} \left[B_1 \cos \frac{\omega x_0}{a} + A_2 \frac{\cos[\omega(l-x_0)/a]}{\sin[\omega l/a]} \right] = A \end{cases}$$

解得

$$\begin{cases} B_1 = \frac{aA}{\omega T} \frac{\sin \frac{\omega(l-x_0)}{a}}{\sin \frac{\omega l}{a}} \\ A_2 = \frac{aA}{\omega T} \sin \frac{\omega x_0}{a} \end{cases}$$

因而, 该稳定解为

$$\begin{cases} u_1(x, t) = \frac{aA}{\omega T} \frac{\sin \frac{\omega(l-x_0)}{a}}{\sin \frac{\omega l}{a}} \sin \frac{\omega x}{a} \cos \omega t, & (0 < x < x_0) \\ u_2(x, t) = \frac{aA}{\omega T} \frac{\sin \frac{\omega x_0}{a}}{\sin \frac{\omega l}{a}} \sin \frac{\omega(l-x)}{a} \cos \omega t, & (x_0 < x < l) \end{cases}$$

$$\begin{cases} B_1 = \frac{aA}{\omega T} \frac{\sin \frac{\omega(l-x_0)}{a}}{\sin \frac{\omega l}{a}} \\ A_2 = \frac{aA}{\omega T} \sin \frac{\omega x_0}{a} \end{cases}$$

因而，该稳定解为

$$\begin{cases} u_1(x,t) = \frac{aA}{\omega T} \frac{\sin \frac{\omega(l-x_0)}{a}}{\sin \frac{\omega l}{a}} \sin \frac{\omega x}{a} \cos \omega t, & (0 < x < x_0) \\ u_2(x,t) = \frac{aA}{\omega T} \frac{\sin \frac{\omega x_0}{a}}{\sin \frac{\omega l}{a}} \sin \frac{\omega(l-x)}{a} \cos \omega t, & (x_0 < x < l) \end{cases}$$

26. 一长为 l 的均匀导热细杆，杆上有热源，单位长度杆上的热源强度为 $c\rho\beta u$, ($\beta > 0$), $x=0$ 端绝热, $x=l$ 端保持摄氏零度, 初始温度分布为 $u_0x(x-l)$, 试求杆上各处温度如何随时间变化的?

其中 c 为杆的比热容, ρ 为杆的线密度, u_0 为常数, 侧面绝热。

解: 定解问题为

$$\begin{aligned} u_t - a^2 u_{xx} &= \beta u \\ u_x|_{x=0} &= 0, \quad u|_{x=l} = 0 \\ u|_{t=0} &= u_0 x(x-l) \end{aligned}$$

设 $u(x, t) = e^{\beta t} w(x, t)$, 将方程齐次化, 有

$$\begin{aligned} w_t - a^2 w_{xx} &= 0 \\ w_x|_{x=0} &= 0, \quad w|_{x=l} = 0 \\ w|_{t=0} &= u_0 x(x-l) \end{aligned}$$

该定解问题的一般解为

$$w(x, t) = \sum_{n=0}^{\infty} C_n e^{-\frac{(2n+1)^2 \pi^2 a^2 t}{4l^2}} \cos \frac{(2n+1)\pi x}{2l}$$

其中,

$$\begin{aligned} C_n &= \frac{2u_0}{l} \int_0^l x(x-l) \cos \frac{(2n+1)\pi x}{2l} dx \\ &= \frac{2u_0}{l} \frac{2l}{(2n+1)\pi} \int_0^l x(x-l) d \left[\sin \frac{(2n+1)\pi x}{2l} \right] \\ &= -\frac{2u_0}{l} \frac{2l}{(2n+1)\pi} \int_0^l (2x-l) \sin \frac{(2n+1)\pi x}{2l} dx \\ &= \frac{2u_0}{l} \left[\frac{2l}{(2n+1)\pi} \right]^2 \int_0^l (2x-l) d \left[\cos \frac{(2n+1)\pi x}{2l} \right] \\ &= \frac{2u_0}{l} \left[\frac{2l}{(2n+1)\pi} \right]^2 \left[(2x-l) \cos \frac{(2n+1)\pi x}{2l} \Big|_0^l - 2 \int_0^l \cos \frac{(2n+1)\pi x}{2l} dx \right] \\ &= \frac{2u_0}{l} \left[\frac{2l}{(2n+1)\pi} \right]^2 \left[l - (-1)^n \frac{4l}{(2n+1)\pi} \right] \end{aligned}$$

因而, 原定解问题的解为

$$\begin{aligned} u(x, t) &= e^{\beta t} \sum_{n=0}^{\infty} \frac{8u_0 l}{(2n+1)^2 \pi^2} \left[l - (-1)^n \frac{4l}{(2n+1)\pi} \right] e^{-\frac{(2n+1)^2 \pi^2 a^2 t}{4l^2}} \cos \frac{(2n+1)\pi x}{2l} \\ &= \sum_{n=0}^{\infty} \frac{2u_0 l^2}{\left(n + \frac{1}{2}\right)^3 \pi^3} \left[\left(n + \frac{1}{2}\right) \pi + 2(-1)^{n+1} \right] e^{\left[\beta - \frac{(n+\frac{1}{2})^2 \pi^2 a^2}{l^2} \right] t} \cos \frac{\left(n + \frac{1}{2}\right) \pi}{l} x. \end{aligned}$$

27. 一圆环形区域，内外环半径分别为 ρ_1 和 ρ_2 ，内环上保持温度为 $u_1 \cos^2 \varphi$ ，外环上保持温度为 $u_2 \sin \varphi$ 。求此圆环区域内的稳定温度分布。

解：类似习题 20，取极坐标，定解问题为

$$\begin{aligned} \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} &= 0, \quad (\rho_1 < \rho < \rho_2) \\ u|_{\rho=\rho_1} &= u_1 \cos^2 \varphi, \quad u|_{\rho=\rho_2} = u_2 \sin \varphi. \end{aligned}$$

本问题的一般解为

$$u(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{n=1}^{\infty} (C_n \rho^n + D_n \rho^{-n}) (A_n \cos n\varphi + B_n \sin n\varphi)$$

内环上的温度为 $u_1 \cos^2 \varphi = \frac{1}{2} u_1 + \frac{1}{2} \cos 2\varphi$ ，该边界条件给出

$$\begin{cases} C_0 + D_0 \ln \rho_1 = \frac{1}{2} u_1 \\ A_2 (C_2 \rho_1^2 + D_2 \rho_1^{-2}) = \frac{1}{2} u_1 \\ A_n (C_n \rho_1^n + D_n \rho_1^{-n}) = 0 \quad (n \neq 1) \\ B_n (C_n \rho_1^n + D_n \rho_1^{-n}) = 0 \end{cases}$$

外环温度 $u_2 \sin \varphi$ 给出

$$\begin{cases} C_0 + D_0 \ln \rho_2 = 0 \\ B_1 (C_1 \rho_2^1 + D_1 \rho_2^{-1}) = u_2 \\ A_n (C_n \rho_2^n + D_n \rho_2^{-n}) = 0 \\ B_n (C_n \rho_1^n + D_n \rho_1^{-n}) = 0 \quad (n \neq 1) \end{cases}$$

上述方程组的解为

$$\begin{cases} C_0 = -\frac{u_1 \ln \rho_2}{2 \ln \frac{\rho_1}{\rho_2}}, \quad D_0 = \frac{u_1}{2 \ln \frac{\rho_1}{\rho_2}} \\ A_2 C_2 = \frac{u_1 \rho_1^2}{2(\rho_1^4 - \rho_2^4)}, \quad A_2 D_2 = -\frac{u_1 \rho_1^2 \rho_2^4}{2(\rho_1^4 - \rho_2^4)} \\ B_1 C_1 = \frac{u_2 \rho_2}{\rho_2^2 - \rho_1^2}, \quad B_1 D_1 = -\frac{u_2 \rho_1^2 \rho_2}{\rho_2^2 - \rho_1^2} \end{cases}$$

其余的系数均为零，因而，解为

$$\begin{aligned} u(\rho, \varphi) &= \frac{u_1 (\ln \rho - \ln \rho_2)}{2 (\ln \rho_1 - \ln \rho_2)} + \left(\frac{u_2 \rho_2}{\rho_2^2 - \rho_1^2} \rho - \frac{u_2 \rho_1^2 \rho_2}{\rho_2^2 - \rho_1^2} \rho^{-1} \right) \sin \varphi \\ &\quad + \left[\frac{u_1 \rho_1^2}{2(\rho_1^4 - \rho_2^4)} \rho^2 - \frac{u_1 \rho_1^2 \rho_2^4}{2(\rho_1^4 - \rho_2^4)} \rho^{-2} \right] \cos 2\varphi \\ &= \frac{u_1}{2} \cdot \frac{\ln \rho_2 - \ln \rho}{\ln \rho_2 - \ln \rho_1} + \frac{u_2 \rho_2}{\rho_2^2 - \rho_1^2} \left(\rho - \frac{\rho_1^2}{\rho} \right) \sin \varphi \\ &\quad - \frac{u_1 \rho_1^2}{2(\rho_2^4 - \rho_1^4)} \left(\rho^2 - \frac{\rho_2^4}{\rho^2} \right) \cos 2\varphi. \end{aligned}$$

28. 一长为 l , 截面积为 S 的均匀细杆, 今将 $x=0$ 端保持为摄氏零度, $x=l$ 端按牛顿冷却定律向温度为零度的媒质散热, 侧面绝热, 原先杆的温度为 u_0 , 求在冷却过程中杆上各处温度的变化。

解: 定解问题为

$$\begin{aligned} u_t - a^2 u_{xx} &= 0 \\ u|_{x=0} &= 0, \quad (hu + ku_x)|_{x=l} = 0 \\ u|_{t=0} &= u_0 \end{aligned}$$

设 $u(x, t) = X(x)T(t)$, 代入定解问题, 分离变数, 有

$$\begin{cases} T' + \lambda a^2 T = 0 \\ X'' + \lambda X = 0 \end{cases}$$

只有 $\lambda > 0$ 满足边界条件, 且

$$\tan \sqrt{\lambda} l = -\frac{k}{h} \sqrt{\lambda}$$

置 $\sqrt{\lambda} l = x$, $\frac{hl}{k} = \eta$, 上式成为

$$x + \eta \tan x = 0$$

h 为热交换系数, k 为热传导系数. 设 $x_1, x_2, x_3, \dots$ 是上述方程的根, 解可以写为

$$u(x, t) = \sum_{n=1}^{\infty} B_n e^{-\frac{x_n^2 a^2 t}{l^2}} \sin \frac{x_n}{l} x,$$

代入初始条件

$$\begin{aligned} u(x, 0) &= \sum_{n=1}^{\infty} B_n \sin \frac{x_n}{l} x = u_0 \\ B_n &= \frac{u_0}{\int_0^l \sin^2 \frac{x_n}{l} x dx} \int_0^l \sin \frac{x_n}{l} x dx \\ &= u_0 \cdot \frac{4 \frac{x_n}{l}}{2x_n - \sin 2x_n} \cdot \frac{1 - \cos x_n}{\frac{x_n}{l}} \\ &= 2u_0 \cdot \frac{1 - \cos x_n}{x_n - \sin x_n \cos x_n} \end{aligned}$$

从而, 解为

$$u(x, t) = \sum_{n=1}^{\infty} \frac{2u_0(1 - \cos x_n)}{x_n - \sin x_n \cos x_n} e^{-\frac{x_n^2 a^2 t}{l^2}} \sin \frac{x_n}{l} x.$$

§ 8.2 非齐次振动方程和输运方程

1. 长为 l 的均匀细杆两端固定, 杆上单位长度受有纵向外力 $f_0 \sin(2\pi x/l) \cos \omega t$, 初始位移为 $[\sin(\pi x/l)]^2$, 初始速度为零, 求解杆的纵振动。

解: 定解问题为

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= \frac{f_0}{ES} \sin \frac{2\pi x}{l} \cos \omega t, \\ u|_{x=0} &= u|_{x=l} = 0, \\ u|_{t=0} &= \sin^2 \frac{\pi x}{l}, \quad u_t|_{t=0} = 0. \end{aligned} \quad (1)$$

设解为

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}.$$

以此解代入定解问题①, 同时, 将初值条件亦按 $\sin \frac{n\pi x}{l}$ 展开,

$$\sin^2 \frac{\pi x}{l} = \frac{1}{2} - \frac{1}{2} \cos \frac{2\pi x}{l} = \sum_{n=1}^{\infty} \varphi_n \sin \frac{n\pi x}{l}.$$

其中,

$$\begin{aligned} \varphi_n &= \frac{2}{l} \int_0^l \left(\frac{1}{2} - \frac{1}{2} \cos \frac{2\pi x}{l} \right) \sin \frac{n\pi x}{l} dx \\ &= \frac{1}{l} \int_0^l \sin \frac{n\pi x}{l} dx - \frac{1}{2l} \int_0^l \left[\sin \frac{(n+2)\pi x}{l} + \sin \frac{(n-2)\pi x}{l} \right] dx \\ &= \frac{1}{n\pi} [1 - (-1)^n] - \frac{1}{2} \left[\frac{1}{(n+2)\pi} + \frac{1}{(n-2)\pi} \right] [1 - (-1)^n] \\ &= \frac{[1 - (-1)^n]}{\pi} \left(\frac{1}{n} - \frac{n}{n^2 - 4} \right) \\ &= -4 \frac{[1 - (-1)^n]}{\pi n (n^2 - 4)} = \begin{cases} -\frac{8}{\pi(2k-1)[(2k-1)^2 - 4]}, & (n = 2k-1, k = 1, 2, 3, \dots) \\ 0, & (n = 2k) \end{cases} \end{aligned}$$

有

$$\begin{cases} T_2'' + \frac{4\pi^2 a^2}{l^2} T_2 = \frac{f_0}{ES} \cos \omega t \\ T_2(0) = 0, \quad T_2'(0) = 0, \end{cases}$$

$$\begin{cases} T_n'' + \frac{n^2 \pi^2 a^2}{l^2} T_n = 0, & (n \neq 2) \\ T_n(0) = \begin{cases} -\frac{8}{\pi(2k-1)[(2k-1)^2-4]}, & (n=2k-1, k=1,2,3\dots), \\ 0, & (n=2k), \end{cases} \\ T_2'(0) = 0. \end{cases} \quad (2)$$

其解为

$$\begin{cases} T_2 = A_2 \cos \frac{2\pi at}{l} + B_2 \sin \frac{2\pi at}{l} + \frac{f_0}{ES} \frac{l^2}{4\pi^2 a^2 - \omega^2 l^2} \cos \omega t \\ T_n = A_n \cos \frac{n\pi at}{l} + B_n \sin \frac{n\pi at}{l} \end{cases} \quad (3)$$

将③代入②中初始条件. 由初始速度为零, 有

$$T_n' \Big|_{t=0} = 0 \Rightarrow B_n = 0 \Rightarrow \begin{cases} T_2 = A_2 \cos \frac{2\pi at}{l} + \frac{f_0}{ES} \frac{l^2}{4\pi^2 a^2 - \omega^2 l^2} \cos \omega t \\ T_n = A_n \cos \frac{n\pi at}{l} \end{cases}$$

初始位移条件给出

$$\begin{cases} T_{2k} \Big|_{t=0} = 0, & n = 2k \\ T_{2k-1} \Big|_{t=0} = -\frac{8}{\pi(2k-1)[(2k-1)^2-4]}, & (n=2k-1, k=1,2,3\dots) \end{cases}$$

有

$$\begin{aligned} A_2 &= -\frac{f_0}{ES} \frac{f_0 l^2}{4\pi^2 a^2 - \omega^2 l^2}, & A_{2k} &= 0 \\ A_{2k-1} &= -\frac{8}{\pi(2k-1)[(2k-1)^2-4]}, & (n=2k-1, k=1,2,3\dots) \end{aligned}$$

因而, 最终所得解为

$$\begin{aligned} u(x,t) &= \sum_{k=1}^{\infty} \frac{-8}{(2k-1)[(2k-1)^2-4]\pi} \cos \frac{(2k-1)\pi a}{l} t \sin \frac{(2k-1)\pi}{l} x \\ &\quad + \frac{f_0}{ES} \cdot \frac{1}{\omega^2 - \frac{4\pi^2 a^2}{l^2}} \left(\cos \frac{2\pi a}{l} t - \cos \omega t \right) \sin \frac{2\pi}{l} x. \end{aligned}$$

2. 长位 l 的均匀细杆受有纵向外力, 求解如下纵振动问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = A \sin \frac{3\pi}{2l} x \cos \omega t, \\ u|_{x=0} = 0, \quad u_x|_{x=l} = 0, \\ u|_{t=0} = u_0 x, \quad u_t|_{t=0} = 0. \end{cases}$$

其中, A 和 u_0 为常数, $0 < x < l$ 。

解: 类似 § 8.1 例 2, u 的本征函数及本征值分别为 $X_n(x) = \sin \sqrt{\lambda_n} x$ 及

$$\lambda_n = \frac{(2n+1)^2 \pi^2}{4l^2}。 \text{ 于是, 可将 } u \text{ 的解设为}$$

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) \sin \frac{(2n+1)\pi x}{2l}。$$

代入定解问题, 得到

$$\begin{cases} \sum_{n=0}^{\infty} \left[T_n''(t) + \frac{(2n+1)^2 \pi^2 a^2}{4l^2} T_n(t) \right] \sin \frac{(2n+1)\pi x}{2l} = A \sin \frac{3\pi x}{2l} \sin \omega t, \\ \sum_{n=0}^{\infty} T_n(0) \sin \frac{(2n+1)\pi x}{2l} = u_0 x, \quad \sum_{n=0}^{\infty} T_n'(0) \sin \frac{(2n+1)\pi x}{2l} = 0. \end{cases}$$

有

$$\begin{aligned} T_1''(t) + \frac{9\pi^2 a^2}{4l^2} T_1(t) &= A \sin \omega t, \\ T_1(0) &= -\frac{8u_0 l}{9\pi^2}, \quad T_1'(0) = 0. \end{aligned}$$

由此解得

$$T_1(t) = \frac{4Al^2}{9\pi^2 a^2 - 4\omega^2 l^2} \cos \omega t - \left(\frac{8u_0 l}{9\pi^2} + \frac{4Al^2}{9\pi^2 a^2 - 4\omega^2 l^2} \right) \cos \frac{3\pi a t}{2l}。$$

$$\begin{cases} T_n''(t) + \frac{(2n+1)^2 \pi^2 a^2}{4l^2} T_n(t) = 0, \\ T_n(0) = -\frac{8u_0 l}{(2n+1)^2 \pi^2}, \\ T_n'(0) = 0. \end{cases} \quad (n=0, 1, 2, \dots)$$

解得

$$T_n(t) = (-1)^n \frac{8u_0 l}{(2n+1)^2 \pi^2} \cos \frac{(2n+1)\pi a t}{2l}。$$

这样, 问题的解是

$$\begin{aligned} u(x, t) &= \left[\frac{4Al^2}{9\pi^2 a^2 - 4\omega^2 l^2} \cos \omega t - \left(\frac{8u_0 l}{9\pi^2} + \frac{4Al^2}{9\pi^2 a^2 - 4\omega^2 l^2} \right) \cos \frac{3\pi a t}{2l} \right] \sin \frac{3\pi x}{2l} \\ &\quad + \sum_{n=0, 2}^{\infty} (-1)^n \frac{8u_0 l}{(2n+1)^2 \pi^2} \cos \frac{(2n+1)\pi a t}{2l} \sin \frac{(2n+1)\pi x}{2l}。 \end{aligned}$$

3. 两端固定的弦，原先静止不动，单位长度所受横向外力为 $\rho f(x,t) = \rho \Phi(x) \sin \omega t$ ，求解弦的振动，研究共振的可能性，并求共振时的解。

解：定解问题为

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= \Phi(x) \sin \omega t \\ u|_{x=0} &= u|_{x=l} = 0 \\ u|_{t=0} &= 0, u_t|_{t=0} = 0 \end{aligned}$$

根据边界条件，设其解为

$$u(x,t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}$$

将 $\Phi(x)$ 亦按 $\sin \frac{n\pi x}{l}$ 展开，

$$\Phi(x) = \sum_{n=1}^{\infty} \varphi_n \sin \frac{n\pi x}{l}, \quad \varphi_n = \frac{2}{l} \int_0^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi,$$

代入定解问题，有

$$T_n''(t) + \frac{n^2 \pi^2 a^2}{l^2} T_n(t) = \varphi_n \sin \omega t.$$

解该常微分方程，有

$$T_n(t) = A_n \cos \frac{n\pi a t}{l} + B_n \sin \frac{n\pi a t}{l} - \frac{\varphi_n}{\omega^2 - \frac{n^2 \pi^2 a^2}{l^2}} \sin \omega t$$

这样，问题的解是

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a t}{l} + B_n \sin \frac{n\pi a t}{l} - \frac{\varphi_n}{\omega^2 - \frac{n^2 \pi^2 a^2}{l^2}} \sin \omega t \right) \sin \frac{n\pi x}{l}.$$

将此解代入初始条件，有

$$\begin{aligned} A_n &= 0 \\ B_n &= \frac{\omega l}{n\pi a} \frac{\varphi_n}{\omega^2 - \frac{n^2 \pi^2 a^2}{l^2}} \end{aligned}$$

结果，问题的解为

$$\begin{aligned} u(x,t) &= \sum_{n=1}^{\infty} \frac{\varphi_n}{\omega^2 - \frac{n^2 \pi^2 a^2}{l^2}} \left(\frac{\omega l}{n\pi a} \sin \frac{n\pi a t}{l} - \sin \omega t \right) \sin \frac{n\pi x}{l} \\ &= \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \int_0^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi \left[\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \right] \sin \frac{n\pi x}{l}. \end{aligned}$$

如外力的频率等于基音或谐音的频率，则[]为 0/0，按罗毕达法则其极限为

$\left(\frac{1}{2\omega} \sin \omega t - \frac{1}{2} t \cos \omega t \right)$ ，其中第二部分的振幅为 $\frac{1}{2}t$ ，随时间而增长，这就是共振。

4. 两端固定弦在点 x_0 受谐变力 $\rho f(t) = \rho f_0 \sin \omega t$ 作用而振动，求解振动情况。[提示：外加力的线密度可表为 $\rho f(x, t) = \rho f_0 \sin \omega t \delta(x - x_0)$]

解：类似第 3 题，定解问题为

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= \rho f_0 \delta(x - x_0) \sin \omega t \\ u|_{x=0} &= u|_{x=l} = 0 \\ u|_{t=0} &= 0, u_t|_{t=0} = 0 \end{aligned}$$

根据边界条件，设其解为

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}。$$

将方程右函数按 $\sin \frac{n\pi x}{l}$ 展开, 其中

$$\delta(x - x_0) = \sum_{n=1}^{\infty} \frac{2}{l} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi x}{l}。$$

这样，类似第 3 题，本题最终解为

$$u(x, t) = \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi x}{l} \left(\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \right)。$$

5.求解振动问题 A 和 B 为常数。

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= Ax, & (0 < x < l) \\ u|_{x=0} &= 0, & u_x|_{x=l} = 0, \\ u|_{t=0} &= 0, & u_t|_{t=0} = Bx. \end{aligned}$$

解：用傅里叶级数展开法求解，类似 § 8.1 例 2， u 的本征函数及本征值分别为分别为

$$X_n(x) = \sin \sqrt{\lambda_n} x$$

及

$$\lambda_n = \frac{(2n+1)^2 \pi^2}{4l^2}.$$

于是，可将 u 的解设为

$$u(x,t) = \sum_{n=0}^{\infty} T_n(t) \sin \frac{(2n+1)\pi x}{2l}.$$

代入定解问题，得到

$$\begin{cases} \sum_{n=0}^{\infty} \left[T_n''(t) + \frac{(2n+1)^2 \pi^2 a^2}{4l^2} T_n(t) \right] \sin \frac{(2n+1)\pi x}{2l} = Ax, \\ \sum_{n=0}^{\infty} T_n(0) \sin \frac{(2n+1)\pi x}{2l} = 0, \\ \sum_{n=0}^{\infty} T_n'(0) \sin \frac{(2n+1)\pi x}{2l} = Bx. \end{cases}$$

有

$$\begin{cases} T_n''(t) + \frac{(2n+1)^2 \pi^2 a^2}{4l^2} T_n(t) = (-1)^n \frac{8lA}{(2n+1)^2 \pi^2}, \\ T_n(0) = 0, \\ T_n'(0) = (-1)^n \frac{8lB}{(2n+1)^2 \pi^2}. \end{cases} \quad (n=0,1,2,\dots)$$

$T_n(t)$ 的解是

$$T_n(t) = (-1)^n \frac{16l^2}{(2n+1)^3 \pi^3 a} \left\{ \frac{2Al}{(2n+1)\pi a} \left[1 - \cos \frac{(2n+1)\pi at}{2l} \right] + B \sin \frac{(2n+1)\pi at}{2l} \right\}.$$

从而，问题的解是

$$\begin{aligned} u(x,t) &= \sum_{n=0}^{\infty} (-1)^n \frac{16l^2}{(2n+1)^3 \pi^3 a} \left\{ \frac{2Al}{(2n+1)\pi a} \left[1 - \cos \frac{(2n+1)\pi at}{2l} \right] + B \sin \frac{(2n+1)\pi at}{2l} \right\} \\ &\quad \times \sin \frac{(2n+1)\pi x}{2l}. \end{aligned}$$

6. 求解输运问题

$$\begin{aligned}u_t - a^2 u_{xx} &= -b u_x, & (0 < x < l) \\ u|_{x=0} &= 0, u|_{x=l} = 0, \\ u|_{t=0} &= \varphi(x).\end{aligned}$$

其中， b 为常数。能否用傅里叶级数法求解？如果不能，要说明原因；如果能，将 $u(x, t)$ 解出来。

解：能用傅里叶级数法求解，设解

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) e^{\frac{b}{2a^2}x} \sin \frac{n\pi}{l} x。$$

代入定解问题，其中

$$\begin{aligned}u_x(x, t) &= \sum_{n=1}^{\infty} T_n(t) \left(\frac{b}{2a^2} e^{\frac{b}{2a^2}x} \sin \frac{n\pi}{l} x + \frac{n\pi}{l} e^{\frac{b}{2a^2}x} \cos \frac{n\pi}{l} x \right) \\ u_{xx}(x, t) &= \sum_{n=1}^{\infty} T_n(t) \left(\frac{b^2}{4a^4} e^{\frac{b}{2a^2}x} \sin \frac{n\pi}{l} x + 2 \cdot \frac{b}{2a^2} \frac{n\pi}{l} e^{\frac{b}{2a^2}x} \cos \frac{n\pi}{l} x - \frac{n^2 \pi^2}{l^2} e^{\frac{b}{2a^2}x} \sin \frac{n\pi}{l} x \right) \\ u_t(x, t) &= \sum_{n=1}^{\infty} T'_n(t) e^{\frac{b}{2a^2}x} \sin \frac{n\pi}{l} x\end{aligned}$$

这样

$$\sum_{n=1}^{\infty} \left[T'_n(t) - \left(\frac{b^2}{4a^2} + \frac{n^2 \pi^2 a^2}{l^2} \right) T_n(t) \right] e^{\frac{b}{2a^2}x} \sin \frac{n\pi}{l} x = 0。$$

于是有

$$T'_n(t) - \left(\frac{b^2}{4a^2} + \frac{n^2 \pi^2 a^2}{l^2} \right) T_n(t) = 0。$$

其解是

$$T_n(t) = \varphi_n e^{-\left(\frac{b^2}{4a^2} + \frac{n^2 \pi^2 a^2}{l^2} \right) t}。$$

于是

$$u(x, t) = \sum_{n=1}^{\infty} \varphi_n e^{-\left(\frac{b^2}{4a^2} + \frac{n^2 \pi^2 a^2}{l^2} \right) t} e^{\frac{b}{2a^2}x} \sin \frac{n\pi}{l} x。$$

代入初值

$$\sum_{n=1}^{\infty} \varphi_n e^{\frac{b}{2a^2}x} \sin \frac{n\pi}{l} x = \varphi(x)。$$

其中 $\varphi_n = \frac{2}{l} \int_0^l \varphi(x) \left(e^{\frac{bx}{2a^2}} \sin \frac{n\pi}{l} x \right) e^{-\frac{bx}{2a^2}} dx$ 。本题中出现的本征函数为 $X_n(x) = A e^{\frac{b}{2a^2}x} \sin \frac{n\pi}{l} x$,

彼此在区间 $[0, l]$ 上带权重函数 $e^{-\frac{b}{a^2}x}$ 正交。

7. 均匀细导线，每单位长的电阻为 R ，通以恒定的电流 I ，导线表面跟周围温度为零度的介质进行热量交换，试解线上温度变化，设初始温度和两端温度都是零度。

解：考虑微元 dx ， dt 时间内该微元所得到的总能量为

$$ku_{xx}dxdt + I^2 R dxdt - hudxdt = c\rho dxdu$$

从而定解问题为

$$\begin{aligned} u_t - a^2 u_{xx} + \frac{h}{c\rho} u &= \frac{1}{c\rho} I^2 R, \\ u|_{x=0} &= u|_{x=l} = 0 \\ u|_{t=0} &= 0 \end{aligned}$$

根据齐次边界条件，设解为

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}.$$

代入方程后有

$$T'_n(t) + \left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) T_n(t) = \left[1 - (-1)^n \right] \frac{2I^2 R}{n\pi c\rho}.$$

解得

$$\begin{aligned} T_n &= C_n e^{-\left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) t} + \frac{\left[1 - (-1)^n \right] \frac{2I^2 R}{n\pi c\rho}}{\left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right)} \\ &= C_n e^{-\left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) t} + \left[1 - (-1)^n \right] \frac{2I^2 R l^2}{n\pi (c\rho n^2 \pi^2 a^2 + h l^2)} \end{aligned}$$

由初始温度为零，有

$$C_n = - \left[1 - (-1)^n \right] \frac{2I^2 R l^2}{n\pi (c\rho n^2 \pi^2 a^2 + h l^2)}.$$

因而

$$T_n(t) = \left[1 - (-1)^n \right] \frac{2I^2 R l^2}{n\pi (c\rho n^2 \pi^2 a^2 + h l^2)} \left[1 - e^{-\left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) t} \right].$$

所以，解为

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} \left[1 - (-1)^n \right] \frac{2I^2 R l^2}{n\pi (c\rho n^2 \pi^2 a^2 + h l^2)} \left[1 - e^{-\left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) t} \right] \sin \frac{n\pi x}{l} \\ &= \frac{4I^2 R}{\pi c\rho} \sum_{k=0}^{\infty} \frac{1}{2k+1} \cdot \frac{1}{(2k+1)^2 \pi^2 a^2 / l^2 + h / c\rho} \left\{ 1 - e^{-\left(\frac{(2k+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) t} \right\} \sin \frac{(2k+1)\pi x}{l} \end{aligned}$$

8. 求解输运问题

$$\begin{aligned} u_t - a^2 u_{xx} &= A \sin \omega t, & (0 < x < l) \\ u|_{x=0} &= 0, & u_x|_{x=l} = 0, \\ u|_{t=0} &= B \cos \frac{\pi}{l} x. \end{aligned}$$

其中， A, B 为常数。

解：类似 § 8.1 例 2， u 的本征函数及本征值分别为 $X_n(x) = \sin \sqrt{\lambda_n} x$ 及 $\lambda_n = \frac{(2n+1)^2 \pi^2}{4l^2}$ ，于是，可将 u 的解设为

$$u(x, t) = \sum_{n=0} T_n(t) \sin \sqrt{\lambda_n} x.$$

代入定解问题，得到

$$\begin{cases} T_n'(t) + \lambda_n a^2 T_n(t) = \frac{2A}{\sqrt{\lambda_n} l} \sin \omega t, \\ T_n(0) = \frac{2B \sqrt{\lambda_n} l}{\lambda_n l^2 - \pi^2}. \end{cases}$$

解得

$$T_n(t) = \left(\frac{2B \sqrt{\lambda_n} l}{\lambda_n l^2 - \pi^2} + \frac{2\omega A}{\sqrt{\lambda_n} l} \frac{1}{\omega^2 + \lambda_n^2 a^4} \right) e^{-\lambda_n a^2 t} + \frac{2A}{\sqrt{\lambda_n} l} \frac{\lambda_n a^2 \sin \omega t - \omega \cos \omega t}{\omega^2 + \lambda_n^2 a^4}.$$

于是，问题的解为

$$\begin{aligned} u(x, t) &= \sum_{n=0}^{\infty} \left[\left(\frac{2A}{\sqrt{\lambda_n} l} \frac{\omega}{\lambda_n^2 a^4 + \omega^2} + \frac{2B \sqrt{\lambda_n} l}{\lambda_n l^2 - \pi^2} \right) e^{-\lambda_n a^2 t} + \frac{2A}{\sqrt{\lambda_n} l} \frac{\lambda_n a^2 \sin \omega t - \omega \cos \omega t}{\lambda_n^2 a^4 + \omega^2} \right] \sin \sqrt{\lambda_n} x, \\ \sqrt{\lambda_n} &= \left(n + \frac{1}{2} \right) \frac{\pi}{l}, \quad n = 0, 1, 2, \dots \end{aligned}$$

§ 8.3 非齐次边界条件的处理

1. 求解细杆导热问题. 杆长 l , 初始温度均匀为 u_0 , 两端分别保持温度 u_1 和 u_2 。

解: 定解问题为

$$\begin{aligned} u_t - a^2 u_{xx} &= 0 \\ u|_{x=0} &= u_1, u|_{x=l} = u_2 \\ u|_{t=0} &= u_0 \end{aligned}$$

令

$$u(x, t) = u_1 + \frac{u_2 - u_1}{l} x + w(x, t)。$$

将边界条件齐次化之后有

$$\begin{aligned} w_t - a^2 w_{xx} &= 0 \\ w|_{x=0} &= 0, w|_{x=l} = 0 \\ w|_{t=0} &= u_0 - u_1 - \frac{u_2 - u_1}{l} x \end{aligned}$$

该定解问题的解为

$$w(x, t) = \sum_{n=1}^{\infty} C_n e^{-\frac{n^2 \pi^2 a^2 t}{l^2}} \sin \frac{n \pi x}{l}。$$

代入初始条件, 确定系数

$$\sum_{n=1}^{\infty} C_n \sin \frac{n \pi x}{l} = u_0 - u_1 - \frac{u_2 - u_1}{l} x。$$

其中

$$\begin{aligned} C_n &= \frac{2}{l} \int_0^l \left(u_0 - u_1 - \frac{u_2 - u_1}{l} x \right) \sin \frac{n \pi x}{l} dx \\ &= \frac{2(u_1 - u_0)}{n \pi} [(-1)^n - 1] + \frac{2(u_2 - u_1)}{n \pi} (-1)^n, \end{aligned}$$

从而, 问题的解为

$$u(x, t) = u_1 + \frac{u_2 - u_1}{l} x + 2 \sum_{n=1}^{\infty} \left\{ \frac{(u_1 - u_0)}{n \pi} [(-1)^n - 1] + \frac{(u_2 - u_1)}{n \pi} (-1)^n \right\} e^{-\frac{n^2 \pi^2 a^2 t}{l^2}} \sin \frac{n \pi x}{l}。$$

2. 求解细杆导热问题，初始温度为零，一端 $x=l$ 保持零度，另一端 $x=0$ 的温度为 At (A 是常数， t 代表时间)。

解：定解问题为

$$\begin{aligned} u_t - a^2 u_{xx} &= 0 \\ u|_{x=0} &= At, \quad u|_{x=l} = 0 \\ u|_{t=0} &= 0 \end{aligned}$$

设

$$u(x,t) = \frac{A}{l}t(l-x) + w(x,t)$$

代入定解问题，将边界条件齐次化，有

$$\begin{aligned} w_t - a^2 w_{xx} &= \frac{A}{l}(x-l) \\ w|_{x=0} &= 0, \quad w|_{x=l} = 0 \\ w|_{t=0} &= 0 \end{aligned}$$

利用傅里叶级数法，设其定解为

$$w(x,t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}$$

代入原方程后有

$$\begin{aligned} T_n'(t) + \frac{n^2\pi^2 a^2}{l^2} T_n(t) &= \frac{2A}{l^2} \int_0^l (x-l) \sin \frac{n\pi x}{l} dx \\ &= -\frac{2A}{l^2} \frac{l}{n\pi} \left[(x-l) \cos \frac{n\pi x}{l} \Big|_0^l - \int_0^l \cos \frac{n\pi x}{l} dx \right] \\ &= -\frac{2A}{n\pi} \end{aligned}$$

解得

$$T_n(t) = C_n e^{-\frac{n^2\pi^2 a^2 t}{l^2}} - \frac{2Al^2}{n^3\pi^3 a^2}$$

于是

$$w(x,t) = \sum_{n=1}^{\infty} \left(C_n e^{-\frac{n^2\pi^2 a^2 t}{l^2}} - \frac{2Al^2}{n^3\pi^3 a^2} \right) \sin \frac{n\pi x}{l}$$

代入初始条件

$$w|_{t=0} = 0 \Rightarrow T_n(0) = 0 \Rightarrow C_n = \frac{2Al^2}{n^3\pi^3 a^2}$$

因而，原定解问题的解为

$$u(x,t) = \frac{A}{l}t(l-x) + \sum_{n=1}^{\infty} \frac{2Al^2}{n^3\pi^3 a^2} \left(e^{-\frac{n^2\pi^2 a^2 t}{l^2}} - 1 \right) \sin \frac{n\pi x}{l}。$$

3.求解均匀杆的纵振动定解问题， A 为常数

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= 0, & (0 < x < l) \\ u|_{x=0} &= 0, & u_x|_{x=l} = A \sin \omega t, \\ u|_{t=0} &= 0, & u_t|_{t=0} = 0. \end{aligned} \tag{①}$$

解：先找一函数 $v(x, t)$ ，使其满足

$$\begin{aligned} v_{tt} - a^2 v_{xx} &= 0, \\ v|_{x=0} &= 0, & v_x|_{x=l} = A \sin \omega t; \end{aligned} \tag{②}$$

令 $v(x, t) = X(x) \sin \omega t$ ，以此代入②，有

$$\begin{cases} X''(x) + \frac{\omega^2}{a^2} X(x) = 0 \\ X(0) = 0, X'(l) = A \end{cases} \Rightarrow X(x) = \frac{Aa}{\omega} \frac{\sin \frac{\omega x}{a}}{\cos \frac{\omega l}{a}}$$

于是有特解

$$v(x, t) = \frac{Aa}{\omega} \frac{\sin \frac{\omega x}{a}}{\cos \frac{\omega l}{a}} \sin \omega t,$$

再令 $u(x, t) = v(x, t) + w(x, t)$ ，以此代入①，有

$$\begin{aligned} w_{tt} - a^2 w_{xx} &= 0, \\ w|_{x=0} &= 0, & w_x|_{x=l} &= 0, \\ w|_{t=0} &= 0, & w_t|_{t=0} &= -Aa \sin \frac{\omega x}{a} \Big/ \cos \frac{\omega l}{a}. \end{aligned}$$

类似 § 8.1 例 2， w 问题的本征函数及本征值分别为 $X_n(x) = \sin \frac{(2n+1)\pi x}{2l}$ 及 $\lambda_n = \frac{(2n+1)^2 \pi^2}{4l^2}$ ， w

问题的一般解是

$$w(x, t) = \sum_{n=0}^{\infty} \left[A_n \cos \frac{(2n+1)\pi at}{2l} + B_n \sin \frac{(2n+1)\pi at}{2l} \right] \sin \frac{(2n+1)\pi x}{2l},$$

其中系数 A_n 和 B_n 可由初值确定，得

$$A_n = 0,$$

$$\begin{aligned}
B_n &= \frac{4}{(2n+1)\pi a} \int_0^l (-Aa) \frac{\sin(\omega \xi / a)}{\cos(\omega l / a)} \sin \frac{(2n+1)\pi \xi}{2l} d\xi \\
&= \frac{-4A}{(2n+1)\pi \cos(\omega l / a)} \left\{ -\frac{\sin[\omega / a + (2n+1)\pi / 2l] \xi}{2[\omega / a + (2n+1)\pi / 2l]} + \frac{\sin[\omega / a - (2n+1)\pi / 2l] \xi}{2[\omega / a - (2n+1)\pi / 2l]} \right\}_0^l \\
&= \frac{2A}{(2n+1)\pi \cos(\omega l / a)} \left\{ \frac{\sin[\omega l / a + (2n+1)\pi / 2]}{\omega / a + (2n+1)\pi / 2l} - \frac{\sin[\omega l / a - (2n+1)\pi / 2]}{\omega / a - (2n+1)\pi / 2l} \right\} \\
&= (-1)^n \frac{2A}{(2n+1)\pi} \left[\frac{1}{\omega / a + (2n+1)\pi / 2l} - \frac{1}{\omega / a - (2n+1)\pi / 2l} \right] \\
&= (-1)^n \frac{4A\omega}{(2n+1)\pi a} \frac{1}{\omega^2 / a^2 - (2n+1)^2 \pi^2 / 4l^4} \\
w(x, t) &= \sum_n (-1)^n \frac{4A\omega}{(2n+1)\pi a} \frac{1}{\omega^2 / a^2 - (2n+1)^2 \pi^2 / 4l^4} \sin \frac{(2n+1)at}{2l} \sin \frac{(2n+1)\pi x}{2l}
\end{aligned}$$

从而,

$$\begin{aligned}
u(x, t) &= \frac{Aa}{\omega} \frac{\sin \frac{\omega x}{a}}{\cos \frac{\omega l}{a}} \sin \omega t \\
&\quad + \sum_n (-1)^n \frac{4A\omega}{(2n+1)\pi a} \frac{1}{\omega^2 / a^2 - (2n+1)^2 \pi^2 / 4l^4} \sin \frac{(2n+1)at}{2l} \sin \frac{(2n+1)\pi x}{2l}。
\end{aligned}$$

4. 求解均匀杆的纵振动问题 (A 为常数)

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= 0, \\ u|_{x=0} &= \cos \frac{\pi a}{l} t, \quad u_x|_{x=l} = 0, \\ u|_{t=0} &= \cos \frac{\pi}{l} x, \quad u_t|_{t=0} = Ax. \quad (0 < x < l) \end{aligned} \quad \textcircled{1}$$

解: 类似第 3 题, 先找一函数 $v(x, t)$, 使其满足

$$\begin{aligned} v_{tt} - a^2 v_{xx} &= 0, \\ v|_{x=0} &= \cos \frac{\pi a}{l} t, \quad v_x|_{x=l} = 0. \end{aligned}$$

令 $v(x, t) = X(x) \cos \frac{\pi a t}{l}$, 代入有

$$\begin{cases} X''(x) + \frac{\pi^2}{l^2} X(x) = 0 \\ X(0) = 1, X'(l) = 0 \end{cases} \Rightarrow X(x) = \cos \frac{\pi x}{l},$$

于是

$$v(x, t) = \cos \frac{\pi x}{l} \cos \frac{\pi a t}{l}.$$

再令 $u(x, t) = v(x, t) + w(x, t)$, 代入①, 有

$$\begin{aligned} w_{tt} - a^2 w_{xx} &= 0, \\ w|_{x=0} &= 0, \quad w_x|_{x=l} = 0, \\ w|_{t=0} &= 0, \quad w_t|_{t=0} = Ax. \end{aligned}$$

类似 § 8.1 例 2, w 问题的本征函数及本征值分别为 $X_n(x) = \sin \frac{(2n+1)\pi x}{2l}$ 及 $\lambda_n = \frac{(2n+1)^2 \pi^2}{4l^2}$, w

问题的一般解是

$$w(x, t) = \sum_{n=0}^{\infty} \left[A_n \cos \frac{(2n+1)\pi a t}{2l} + B_n \sin \frac{(2n+1)\pi a t}{2l} \right] \sin \frac{(2n+1)\pi x}{2l},$$

其中系数 A_n 和 B_n 可由初值确定, 有

$$\begin{aligned} A_n &= 0, \\ B_n &= \frac{4A}{(2n+1)\pi a} \int_0^l \xi \sin \frac{(2n+1)\pi \xi}{2l} d\xi \\ &= (-1)^n \frac{16Al^2}{(2n+1)^3 \pi^3 a} \end{aligned}$$

于是

$$w(x, t) = \sum_{n=0}^{\infty} (-1)^n \frac{16Al^2}{(2n+1)^3 \pi^3 a} \sin \frac{(2n+1)\pi a t}{2l} \sin \frac{(2n+1)\pi x}{2l}.$$

从而, u 的解是

$$u(x, t) = \cos \frac{\pi x}{l} \cos \frac{\pi a t}{l} + \sum_{n=0}^{\infty} (-1)^n \frac{16Al^2}{(2n+1)^3 \pi^3 a} \sin \frac{(2n+1)\pi a t}{2l} \sin \frac{(2n+1)\pi x}{2l}.$$

5. 长为 l 的柱形管，一端封闭，另一端开放. 管外空气中含有某种气体，其浓度为 u_0 ，向管内扩散。求解该气体在管内的浓度 $u(x, t)$ （图 8-8）。

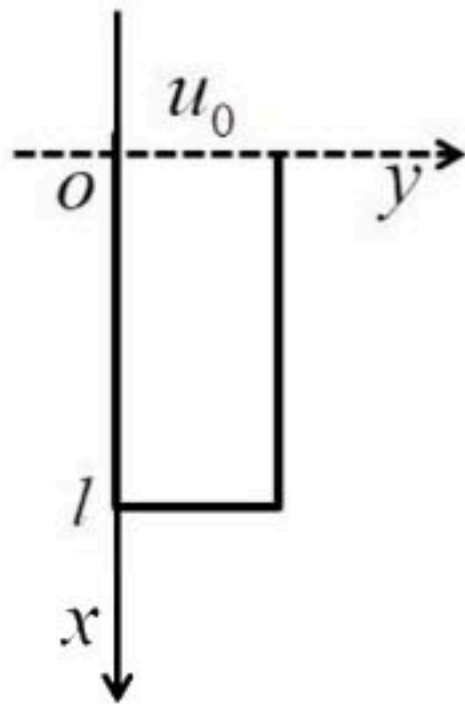


图8-8

解：定解问题为

$$\begin{aligned} u_t - a^2 u_{xx} &= 0 \\ u|_{x=0} &= u_0, \quad u_x|_{x=l} = 0 \\ u|_{t=0} &= 0 \end{aligned}$$

令

$$u(x, t) = u_0 + w(x, t)。$$

以此代入定解问题，得到

$$\begin{aligned} w_t - a^2 w_{xx} &= 0 \\ w|_{x=0} &= 0, \quad w_x|_{x=l} = 0 \\ w|_{t=0} &= -u_0 \end{aligned}$$

类似 § 8.1 例 2， w 的一般解为

$$w(x, t) = \sum_{n=1}^{\infty} C_n e^{-\frac{(n+1/2)^2 \pi^2 a^2 t}{l^2}} \sin \frac{(n+1/2)\pi x}{l}$$

代入初值

$$w|_{t=0} = \sum_{n=1}^{\infty} C_n \sin \frac{(n+1/2)\pi x}{l} = -u_0$$

其中

$$\begin{aligned} C_n &= \frac{2(-u_0)}{l} \int_0^l \sin \frac{(n+1/2)\pi x}{l} dx \\ &= \frac{2u_0}{l} \frac{l}{(n+1/2)\pi} \cos \frac{(n+1/2)\pi x}{l} \Big|_0^l \\ &= -\frac{4u_0}{\pi} \frac{1}{(2n+1)}。 \end{aligned}$$

从而，原定解问题的定解为

$$u(x, t) = u_0 - \frac{4u_0}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} e^{-(2n+1)^2 \pi^2 a^2 t/4l^2} \sin \frac{(2n+1)\pi x}{2l}。$$

6. 把弹簧上端 $x=0$ 加以固定, 在静止弹簧的下端 $x=l$ 轻轻地挂上质量为 m 的物体, 求解弹簧的纵振动。弹簧本身的重量可忽略不计。

解: 定解问题为

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= 0, \\ u|_{x=0} &= 0, \\ mg - ES u_x|_{x=l} &= mu_{tt}|_{x=l}, \\ \text{即 } \left(u_x + \frac{m}{ES} u_{tt} \right) \Big|_{x=l} &= \frac{mg}{ES}, \\ u|_{t=0} &= 0, \quad u_t|_{t=0} = 0. \end{aligned} \tag{1}$$

令 $u(x,t) = w(x,t) + \frac{mg}{ES}x$ 代入①后, 有

$$\begin{aligned} w_{tt} - a^2 w_{xx} &= 0, \\ w|_{x=0} &= 0, \\ \left(w_x + \frac{m}{ES} w_{tt} \right) \Big|_{x=l} &= 0, \\ w|_{t=0} &= -\frac{mg}{ES}x, \quad w_t|_{t=0} = 0, \end{aligned} \tag{2}$$

令 $w(x,t) = X(x)T(t)$, 代入定解问题②, 分离变数后的常微分方程为

$$X'' + \lambda X = 0, \tag{3}$$

$$X(0) = 0, \tag{4}$$

$$X'(l)T(t) + \frac{m}{ES} X(l)T''(t) = 0, \tag{5}$$

$$T'' + \lambda a^2 T = 0, \tag{6}$$

由⑤ ⑥可得出

$$X'(l) - \frac{\lambda m}{\rho S} X(l) = 0. \tag{7}$$

由③ ④⑦求解 X 。当 $\lambda > 0$, 有解

$$X(x) = A \cos \sqrt{\lambda}x + B \sin \sqrt{\lambda}x$$

代入边界条件④, 有 $A = 0$, $\Rightarrow X(x) = B \sin \sqrt{\lambda}x$, 由⑦有

$$\sqrt{\lambda} \cos \sqrt{\lambda}l - \frac{\lambda m}{\rho S} \sin \sqrt{\lambda}l = 0, \tag{8}$$

即

$$\sqrt{\lambda} \tan \sqrt{\lambda} l = \frac{\rho S}{m},$$

记 $l\sqrt{\lambda} \equiv \xi$, ⑧ 可写成 $\tan \xi = \frac{\rho S l}{m} \frac{1}{\xi}$, 可用图解法求解此方程, 设方程⑧的正根为

$\lambda_1, \lambda_2, \lambda_3, \dots$, 则, 方程③的本征函数为

$$X_n(x) = B_n \sin \sqrt{\lambda_n} x, \quad n = 1, 2, 3, \dots$$

这样, ②的一般解为

$$w(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \sqrt{\lambda_n} at + B_n \sin \sqrt{\lambda_n} at \right) \sin \sqrt{\lambda_n} x, \quad (9)$$

由于初始速度为零, 有 $B_n = 0$ 。利用初始位移条件

$$\sum_{n=1}^{\infty} A_n \sin \sqrt{\lambda_n} x = -\frac{mg}{ES} x \equiv \varphi(x), \quad (10)$$

为求系数 A_k , 等式两边同乘 $\sin \sqrt{\lambda_k} x$, 在 $[0, l]$ 上积分

$$\sum_{n=1}^{\infty} A_n \int_0^l \sin \sqrt{\lambda_n} x \sin \sqrt{\lambda_k} x dx = \int_0^l \varphi(x) \sin \sqrt{\lambda_k} x dx.$$

其中, 当 $n \neq k$ 时,

$$\begin{aligned} & \int_0^l \sin \sqrt{\lambda_n} x \sin \sqrt{\lambda_k} x dx \\ &= \frac{1}{2} \left[\int_0^l \cos(\sqrt{\lambda_n} - \sqrt{\lambda_k}) x dx - \int_0^l \cos(\sqrt{\lambda_n} + \sqrt{\lambda_k}) x dx \right] \\ &= \frac{1}{2} \left[\frac{\sin(\sqrt{\lambda_n} - \sqrt{\lambda_k}) l}{\sqrt{\lambda_n} - \sqrt{\lambda_k}} - \frac{\sin(\sqrt{\lambda_n} + \sqrt{\lambda_k}) l}{\sqrt{\lambda_n} + \sqrt{\lambda_k}} \right] \\ &= \frac{1}{2} \left(\frac{\sqrt{\lambda_n} \sin \sqrt{\lambda_n} l \cos \sqrt{\lambda_k} l - \sqrt{\lambda_k} \cos \sqrt{\lambda_k} l \sin \sqrt{\lambda_n} l}{\lambda_k - \lambda_n} \right) \\ &= -\frac{m}{\rho S} \sin \sqrt{\lambda_k} l \sin \sqrt{\lambda_n} l, \quad (\text{由⑧式得到}) \end{aligned}$$

当 $n = k$ 时,

$$\begin{aligned}
& \int_0^l \sin^2 \sqrt{\lambda_k} x dx \\
&= \frac{1}{2} \int_0^l (1 - \cos 2\sqrt{\lambda_k} x) dx \\
&= \frac{l}{2} - \frac{1}{4\sqrt{\lambda_k}} \sin 2\sqrt{\lambda_k} x \Big|_0^l \\
&= \frac{l}{2} - \frac{1}{2\sqrt{\lambda_k}} \sin \sqrt{\lambda_k} l \cos \sqrt{\lambda_k} l
\end{aligned}$$

综合前面几式，有

$$\begin{aligned}
\int_0^l \varphi(x) \sin \sqrt{\lambda_k} x dx &= \sum_{n=1}^{\infty} A_n \int_0^l \sin \sqrt{\lambda_n} x \sin \sqrt{\lambda_k} x dx \\
&= A_k \left[\frac{l}{2} - \frac{m}{2\rho S} \sin^2 \sqrt{\lambda_k} l \right] + \sum_{n \neq k} \left(-\frac{m}{\rho S} \right) A_n \sin \sqrt{\lambda_n} l \sin \sqrt{\lambda_k} l,
\end{aligned}$$

而

$$\frac{m}{\rho S} \varphi(l) \sin \sqrt{\lambda_k} l = \sum_{n=1}^{\infty} \frac{m}{\rho S} A_n \sin \sqrt{\lambda_n} l \sin \sqrt{\lambda_k} l,$$

将上述两式两边相加，得

$$\begin{aligned}
& \int_0^l \varphi(x) \sin \sqrt{\lambda_k} x dx + \frac{m}{\rho S} \varphi(l) \sin \sqrt{\lambda_k} l \\
&= A_k \left(\frac{l}{2} - \frac{m}{2\rho S} \sin^2 \sqrt{\lambda_k} l \right) + \frac{m}{\rho S} A_k \sin^2 \sqrt{\lambda_k} l \\
&= A_k \left(\frac{l}{2} + \frac{m}{2\rho S} \sin^2 \sqrt{\lambda_k} l \right),
\end{aligned}$$

所以

$$\begin{aligned}
\int_0^l \varphi(x) \sin \sqrt{\lambda_k} x dx &= \int_0^l -\frac{mg}{ES} x \sin \sqrt{\lambda_k} x dx \\
&= \frac{mg}{ES} \frac{1}{\lambda_k} \left[\sqrt{\lambda_k} l \cos \sqrt{\lambda_k} l - \sin \sqrt{\lambda_k} l \right] \\
\text{或} \quad &= \frac{mg}{ES} \frac{1}{\lambda_k} \left(\frac{\lambda_k m}{\rho S} l - 1 \right) \sin \sqrt{\lambda_k} l.
\end{aligned}$$

又有

$$\begin{aligned}
\frac{m}{\rho S} \varphi(l) \sin \sqrt{\lambda_k} l &= \frac{m}{\rho S} \left(-\frac{mg}{ES} l \right) \sin \sqrt{\lambda_k} l \\
\text{或} \quad &= -\frac{m^2 g l}{E \rho S^2} \sin \sqrt{\lambda_k} l,
\end{aligned}$$

所以，前式左端为

$$-\frac{mg}{ES\lambda_n}\sin\sqrt{\lambda_n}l$$

$$\text{或} = \frac{mg}{ES}\frac{1}{\lambda_k}\left(\sqrt{\lambda_k}l\cos\sqrt{\lambda_k}l-\sin\sqrt{\lambda_k}l\right)-\frac{m^2gl}{E\rho S^2}\sin\sqrt{\lambda_k}l,$$

从而

$$A_k = \frac{-\frac{2mg}{ES\lambda_k}\sin\sqrt{\lambda_k}l}{l+\frac{m}{\rho S}\sin^2\sqrt{\lambda_k}l}$$

$$\text{或} = \frac{\frac{2mg}{ES}\frac{1}{\lambda_k}\left(\sqrt{\lambda_k}l\cos\sqrt{\lambda_k}l-\sin\sqrt{\lambda_k}l\right)-\frac{m^2gl}{E\rho S^2}\sin\sqrt{\lambda_k}l}{l+\frac{m}{\rho S}\sin^2\sqrt{\lambda_k}l}$$

最后，得本问题的解

$$u(x,t) = \frac{mg}{ES}x + \sum_{k=1}^{\infty} \frac{-2mg\sin\sqrt{\lambda_k}l}{ES\lambda_k\left(l+\frac{m}{\rho S}\sin^2\sqrt{\lambda_k}l\right)}\cos\sqrt{\lambda_k}at\sin\sqrt{\lambda_k}x$$

$$\text{或} = \frac{mg}{ES}x + \sum_{k=1}^{\infty} \frac{\frac{2mg}{ES}\frac{1}{\lambda_k}\left(\sqrt{\lambda_k}l\cos\sqrt{\lambda_k}l-\sin\sqrt{\lambda_k}l\right)-\frac{m^2gl}{E\rho S^2}\sin\sqrt{\lambda_k}l}{l+\frac{m}{\rho S}\sin^2\sqrt{\lambda_k}l}\cos\sqrt{\lambda_k}at\sin\sqrt{\lambda_k}x,$$

又由⑧式有

$$\sin^2\sqrt{\lambda_k}l = \frac{1}{1+\cot\sqrt{\lambda_k}l} = \frac{1}{1+\frac{m^2\lambda_k}{\rho^2S^2}} = \frac{\rho^2S^2}{\rho^2S^2+m^2\lambda_k},$$

从而，本问题的解也可写成

$$u(x,t) = \frac{mg}{ES}x + \sum_{k=1}^{\infty} \frac{-2mg\rho}{E\lambda_k\left(l+\frac{\rho Sm}{\rho^2S^2+m^2\lambda_k}\right)\sqrt{\rho^2S^2+m^2\lambda_k}}\cos\sqrt{\lambda_k}at\sin\sqrt{\lambda_k}x,$$

或者

$$u(x,t) = \frac{mg}{YS}x + \sum_{k=1}^{\infty} \frac{2}{l+\frac{\rho Sm}{\rho^2S^2+m^2\lambda_k}} \left[\frac{mg}{YS\lambda_k}\left(\sqrt{\lambda_k}l\cos\sqrt{\lambda_k}l\right. \right. \\ \left. \left. -\sin\sqrt{\lambda_k}l\right)-\frac{m^2g}{\rho ES^2}l\sin\sqrt{\lambda_k}l \right] \sin\sqrt{\lambda_k}x\cos a\sqrt{\lambda_k}t.$$

其中 λ_n 是 $\sqrt{\lambda}\tan\sqrt{\lambda}l = \frac{S\rho}{m}$ 的第 n 个根。注意，本题的本征函数 $\sin\sqrt{\lambda_n}x$ 并不彼此正交，即

$$\int_0^l \sin\sqrt{\lambda_m}x\sin\sqrt{\lambda_n}xdx \neq 0, \text{ 而是 } \int_0^l \sin\sqrt{\lambda_m}x\sin\sqrt{\lambda_n}xdx + \frac{m}{S\rho}\sin\sqrt{\lambda_m}l\sin\sqrt{\lambda_n}l = 0.$$

§ 8.4 泊松方程

1. 在圆域 $\rho < \rho_0$ 上求解 $\Delta u = -4$, 边界条件是 $u|_{\rho=\rho_0} = 0$ 。

解: 定解问题为

$$\begin{aligned} \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} &= -4, \\ u|_{\rho=\rho_0} &= 0. \end{aligned}$$

令 $u = w - \rho^2$, 代入定解问题, 有

$$\begin{aligned} \frac{\partial^2 w}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial w}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 w}{\partial \varphi^2} &= 0, \\ w|_{\rho=\rho_0} &= \rho_0^2. \end{aligned}$$

其一般解是

$$w(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{n=1}^{\infty} (C_n \rho^n + D_n \rho^{-n}) (A_n \cos n\varphi + B_n \sin n\varphi).$$

代入边界条件 $C_0 + D_0 \ln \rho_0 + \sum_{n=1}^{\infty} (C_n \rho_0^n + D_n \rho_0^{-n}) (A_n \cos n\varphi + B_n \sin n\varphi) = \rho_0^2$ 。两边比较系数, 问

题与 φ 无关, 且仅 $C_2 = 1$ 。从而, w 问题的解为

$$w = \rho_0^2,$$

因而, 原泊松方程的解为

$$u(\rho, \varphi) = \rho_0^2 - \rho^2.$$

2. 在圆域 $\rho < \rho_0$ 上求解泊松方程问题: $\Delta u = 32x^2y$, 边界条件是 $u|_{\rho=\rho_0} = 0$ 。

解: 定解问题是

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} = 32x^2y,$$
$$u|_{\rho=\rho_0} = 0.$$

方程右端

$$32x^2y = 32\rho^3 \cos^2 \theta \sin \theta = 16\rho^3 (\sin 3\varphi - \sin \varphi \cos 2\varphi)$$
$$= 16\rho^3 (\sin \varphi + \sin \varphi \cos 2\varphi)$$

同时又

由此两式相加可得

$$32x^2y = 32\rho^3 \cos^2 \theta \sin \theta = 8\rho^3 \sin 3\varphi + 8\rho^3 \sin \varphi.$$

从而, 可设 u 的特解为 $v(\rho, \varphi) = A\rho^5 \sin \varphi + B\rho^5 \sin 3\varphi$, 代入定解问题, 仅要求其满足方

程, 可确定出系数 $A = \frac{1}{3}$, $B = \frac{1}{2}$ 。

于是, 特解为

$$v(\rho, \varphi) = \frac{1}{3}\rho^5 \sin \varphi + \frac{1}{2}\rho^5 \sin 3\varphi.$$

再令 $u(\rho, \varphi) = v(\rho, \varphi) + w(\rho, \varphi)$, 代入定解问题, 得到 w 的定解问题

$$\begin{cases} \frac{\partial^2 w}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial w}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 w}{\partial \varphi^2} = 0, \\ w|_{\rho=\rho_0} = -\frac{1}{3}\rho_0^5 \sin \varphi - \frac{1}{2}\rho_0^5 \sin 3\varphi. \end{cases}$$

w 的一般解是

$$w(\rho, \varphi) = C_0 + \sum_{m=1}^{\infty} \rho^m (C_m \cos m\varphi + D_m \sin m\varphi).$$

代入边值, 所有 $C_m = 0$, 仅 $D_1 = -\frac{1}{3}\rho_0^4$, $D_3 = -\frac{1}{3}\rho_0^2$.

从而 $w(\rho, \varphi) = -\frac{1}{3}\rho_0^4 \rho \sin \varphi - \frac{1}{2}\rho_0^2 \rho^3 \sin 3\varphi$

结果, 原定解问题的解是

$$u(\rho, \varphi) = \frac{1}{3}\rho(\rho^4 - \rho_0^4) \sin \varphi + \frac{1}{2}\rho^3(\rho^2 - \rho_0^2) \sin 3\varphi.$$

3. 在矩形域 $0 < x < a, -b/2 < y < +b/2$ 上求解 $\Delta u = -2$ ，且 u 在边界上的值为零。

解：定解问题为

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= -2, \\ u|_{x=0} &= u|_{x=a} = 0, \\ u|_{y=-b/2} &= u|_{y=b/2} = 0, \end{aligned} \tag{1}$$

先找一特解 v ，使满足

$$\begin{cases} \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = -2, \\ v|_{x=0} = v|_{x=a} = 0, \end{cases}$$

以试探解形式 $v = A + Bx + Cx^2$ ，代入，求得 $v = ax - x^2$ 。

再设 $u(x, y) = w(x, y) - x(x - a)$ ，以此代入定解问题①，得到

$$\begin{aligned} \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} &= 0 \\ w|_{x=0} &= 0, \quad w|_{x=a} = 0, \\ w|_{y=-b/2} &= x(x - a), \quad w|_{y=b/2} = x(x - a). \end{aligned} \tag{2}$$

②的一般解为

$$w(x, y) = \sum_{n=1}^{\infty} \left(A_n e^{\frac{n\pi y}{a}} + B_n e^{-\frac{n\pi y}{a}} \right) \sin \frac{n\pi x}{a},$$

以此代入 y 方向边值，

$$\begin{aligned} \sum_{n=1}^{\infty} \left(A_n e^{\frac{n\pi b}{2a}} + B_n e^{-\frac{n\pi b}{2a}} \right) \sin \frac{n\pi x}{a} &= x(x - a), \\ \sum_{n=1}^{\infty} \left(A_n e^{\frac{n\pi b}{2a}} + B_n e^{-\frac{n\pi b}{2a}} \right) \sin \frac{n\pi x}{a} &= x(x - a), \end{aligned}$$

有

$$\begin{aligned} A_n e^{\frac{n\pi b}{2a}} + B_n e^{-\frac{n\pi b}{2a}} &= \frac{2}{a} \int_0^a (x^2 - ax) \sin \frac{n\pi x}{a} dx, \\ A_n e^{\frac{n\pi b}{2a}} + B_n e^{-\frac{n\pi b}{2a}} &= \frac{2}{a} \int_0^a (x^2 - ax) \sin \frac{n\pi x}{a} dx, \end{aligned} \tag{3}$$

其中

$$\begin{aligned}\int_0^a x \sin \frac{n\pi x}{a} dx &= \frac{-a}{n\pi} x \sin \frac{n\pi x}{a} \Big|_0^a + \frac{a}{n\pi} \int_0^a \cos \frac{n\pi x}{a} dx, \\ &= \frac{-a^2}{n\pi} (-1)^n + \frac{a}{n\pi} \frac{a}{n\pi} \sin \frac{n\pi x}{a} \Big|_0^a \\ &= \frac{a^2}{n\pi} (-1)^{n+1}\end{aligned}$$

$$\begin{aligned}\int_0^a x^2 \sin \frac{n\pi x}{a} dx &= \frac{-a}{n\pi} x^2 \cos \frac{n\pi x}{a} \Big|_0^a + \frac{2a}{n\pi} \int_0^a x \cos \frac{n\pi x}{a} dx, \\ &= \frac{-a^3}{n\pi} (-1)^n + \frac{2a}{n\pi} x \frac{a}{n\pi} \sin \frac{n\pi x}{a} \Big|_0^a - \frac{2a}{n\pi} \frac{a}{n\pi} \int_0^a \sin \frac{n\pi x}{a} dx \\ &= \frac{-a^3}{n\pi} (-1)^n + \frac{2a^2}{n^2 \pi^2} \frac{a}{n\pi} \cos \frac{n\pi x}{a} \Big|_0^a \\ &= \frac{a^3}{n\pi} (-1)^{n+1} + \frac{2a^3}{n^3 \pi^3} [(-1)^n - 1]\end{aligned}$$

从而

$$\frac{2}{a} \int_0^a (x^2 - ax) \sin \frac{n\pi x}{a} dx = \frac{4a^2}{n^3 \pi^3} [(-1)^n - 1],$$

这样, 由③解出

$$A_n = B_n = \frac{2a^2}{n^3 \pi^3 \operatorname{ch} \frac{n\pi b}{2a}} [(-1)^n - 1].$$

从而, 问题的解为

$$u(x, y) = x(a-x) - \frac{8a^2}{\pi^3} \sum_{n=0}^{\infty} \frac{\operatorname{ch}[(2n+1)\pi y/a] \sin[(2n+1)\pi x/a]}{(2n+1)^3 \operatorname{ch} \left[\left(n + \frac{1}{2} \right) \pi b/a \right]}.$$

4. 在矩形区域 $0 < x < a, 0 < y < b$ 上求解 $\Delta u = -6xy$, 且 u 在边界上的值为零.

解: 定解问题为

$$\begin{aligned} u_{xx} + u_{yy} &= -6xy, \\ u|_{x=0} &= u|_{x=a} = 0, \\ u|_{y=0} &= u|_{y=b} = 0. \end{aligned}$$

取特解 $v(x, y) = (a^2 - x^2)xy$, 其满足

$$\begin{cases} v_{xx} + v_{yy} = -6xy, \\ v|_{x=0} = v|_{x=a} = 0. \end{cases}$$

再令 $u(x, y) = v(x, y) + w(x, y)$, 代入定解问题, 有

$$\begin{aligned} w_{xx} + w_{yy} &= 0, \\ w|_{x=0} &= w|_{x=a} = 0, \\ w|_{y=0} &= 0, w|_{y=b} = (x^2 - a^2)xb. \end{aligned}$$

w 的一般解是

$$w(x, y) = \sum_{n=1}^{\infty} (A_n e^{\frac{n\pi y}{a}} + B_n e^{-\frac{n\pi y}{a}}) \sin \frac{n\pi}{a} x.$$

代入 y 方向边值,

$$\begin{cases} \sum_{n=1}^{\infty} (A_n + B_n) \sin \frac{n\pi}{a} x = 0, \\ \sum_{n=1}^{\infty} (A_n e^{\frac{n\pi b}{a}} + B_n e^{-\frac{n\pi b}{a}}) \sin \frac{n\pi}{a} x = bx(x^2 - a^2). \end{cases}$$

有

$$\begin{cases} A_n + B_n = 0 \\ A_n e^{\frac{n\pi b}{a}} + B_n e^{-\frac{n\pi b}{a}} = (-1)^n \frac{12a^3b}{n^3\pi^3} \end{cases}$$

解得

$$A_n = -B_n = (-1)^n \frac{12a^3b}{n^3\pi^3} \frac{1}{2\text{sh}(n\pi b/a)}.$$

于是

$$w(x, y) = \sum_{n=1}^{\infty} (-1)^n \frac{12a^3b}{n^3\pi^3} \frac{\text{sh } n\pi y/a}{\text{sh } n\pi b/a} \sin \frac{n\pi x}{a}.$$

从而, u 的解是

$$u(x, y) = (a^2 - x^2)xy + \sum_{n=1}^{\infty} (-1)^n \frac{12a^3b}{n^3\pi^3} \frac{\text{sh } n\pi y/a}{\text{sh } n\pi b/a} \sin \frac{n\pi x}{a}.$$